

# Chapter 8

## Parameter List

### 8.1 Parameters (2000h/C00)

| Index | Parameter | Name                                  | Options                                                                                                                                                                                                                                                                  | Value Range   | Default | Unit                | Data Type | Modification Mode | Effective Time   |
|-------|-----------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|---------------------|-----------|-------------------|------------------|
| 01h   | C00.00    | Control mode                          | 0: Position mode<br>1: Speed mode<br>2: Torque mode<br>3: Position → Speed<br>4: Position → Torque<br>5: Speed → Torque<br>6: Position → Speed → Torque                                                                                                                  | 0-6           | 0       | -                   | U16       | At stop           | Immediately      |
| 02h   | C00.01    | Motor rotating direction              | 0: CCW<br>1: CW                                                                                                                                                                                                                                                          | 0-1           | 0       | -                   | U16       | At stop           | Upon re-power-on |
| 03h   | C00.02    | Number of pulses in one turn by motor | -                                                                                                                                                                                                                                                                        | $0(2^{32}-1)$ | 10000   | Unit in application | U32       | At stop           | Upon re-power-on |
| 05h   | C00.04    | Auto-tuning mode                      | 0: Manual mode<br>1: Standard mode<br>2: Positioning mode                                                                                                                                                                                                                | 0-2           | 1       | -                   | U16       | During operation  | Immediately      |
| 06h   | C00.05    | Stiffness level                       | -                                                                                                                                                                                                                                                                        | 1-31          | 12      | -                   | U16       | During operation  | Immediately      |
| 07h   | C00.06    | Load inertia ratio                    | -                                                                                                                                                                                                                                                                        | 0-12000       | 100     | %                   | U16       | During operation  | Immediately      |
| 08h   | C00.07    | Absolute mode                         | 0: Incremental position mode<br>1: Absolute position linear mode<br>2: Absolute position linear infinite mode<br>3: Absolute position single-turn mode<br>4: Absolute position rotation mode<br>5: Absolute mechanical single-turn mode (operating direction selectable) | 0-5           | 0       | -                   | U16       | At stop           | Upon re-power-on |

| Index | Parameter | Name                                                  | Options                                                                                                                        | Value Range | Default | Unit     | Data Type | Modification Mode | Effective Time   |
|-------|-----------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------|---------|----------|-----------|-------------------|------------------|
| 11h   | C00.10    | Bleeder resistor selection                            | 0: Internal bleeder resistor<br>1: External bleeder resistor<br>2: No bleeder resistor<br>3: Capacitor bleeder resistor        | 0-3         | 0       | -        | U16       | At stop           | Immediately      |
| 12h   | C00.11    | Bleeder resistor power                                | -                                                                                                                              | 1-65535     | 50      | W        | U16       | At stop           | Immediately      |
| 13h   | C00.12    | Bleeder resistor resistance                           | -                                                                                                                              | 1-65535     | 50      | $\Omega$ | U16       | At stop           | Immediately      |
| 14h   | C00.13    | Bleeder resistor heat dissipation coefficient         | -                                                                                                                              | 1-100       | 30      | -        | U16       | During operation  | Immediately      |
| 15h   | C00.14    | Brake enable switch                                   | -                                                                                                                              | 0-1         | 0       | -        | U16       | At stop           | Immediately      |
| 17h   | C00.16    | Panel display                                         | 0: Default display<br>1: Speed display<br>2: Torque display<br>3: Voltage display<br>4: Load rate display                      | 0-4         | 0       | -        | U16       | During operation  | Immediately      |
| 18h   | C00.17    | Fully closed-loop disconnection detection             | 0: Shielding detection<br>1: Detect phases AB only<br>2: Detect phases ABZ                                                     | 0-2         | 1       | -        | U16       | At stop           | Upon re-power-on |
| 19h   | C00.18    | Fully closed-loop disconnection detection filter time | -                                                                                                                              | 0-6000      | 400     | ns       | U16       | At stop           | Upon re-power-on |
| 1Ah   | C00.19    | Fully closed-loop hardware filter time                | -                                                                                                                              | 0-6000      | 125     | ns       | U16       | At stop           | Upon re-power-on |
| 18h   | C00.1A    | ABZ port selection                                    | 0: Frequency-division output<br>1: Position comparison output<br>2: Fully closed-loop input<br>3: Pulse synchronization output | 0-3         | 0       | -        | U16       | At stop           | Upon re-power-on |

| Index | Parameter | Name                                       | Options                                                                                                                                                                                                                                              | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|------------------|
| 1Ch   | C00.1B    | OCZ port selection                         | 0: Z-phase OC output<br>1: position comparison output.<br>2: Pulse synchronous output                                                                                                                                                                | 0-2         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 21h   | C00.20    | Pulse input mode selection                 | 0: Pulse+direction, positive logic<br>1: Pulse+direction, negative logic<br>2: AB orthogonal, quadruple frequency<br>3: CW+CCW, positive logic<br>4: CW+CCW, negative logic                                                                          | 0-4         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 22h   | C00.21    | Pulse active edge selection                | 0: Falling edge<br>1: Rising edge                                                                                                                                                                                                                    | 0-1         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 23h   | C00.22    | Pulse channel selection                    | 0: Normal channel<br>1: High-speed channel                                                                                                                                                                                                           | 0-1         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 25h   | C00.24    | Low-speed pulse input filter time          | -                                                                                                                                                                                                                                                    | 0-6000      | 750     | ns   | U16       | At stop           | Upon re-power-on |
| 26h   | C00.25    | High-speed pulse input filter time         | -                                                                                                                                                                                                                                                    | 0-6000      | 75      | ns   | U16       | At stop           | Upon re-power-on |
| 28h   | C00.27    | Frequency-division output phase            | 0: Phase A leads phase B<br>1: Phase B leads phase A                                                                                                                                                                                                 | 0-1         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 29h   | C00.28    | Number of frequency-division output pulses | -                                                                                                                                                                                                                                                    | 0-4194304   | 2500    | P    | U32       | At stop           | Upon re-power-on |
| 28h   | C00.2A    | Z-pulse output polarity selection          | 0: Differential Z positive polarity, OCZ positive polarity<br>1: Differential Z positive polarity, OCZ negative polarity<br>2: Differential Z negative polarity, OCZ positive polarity<br>3: Differential Z negative polarity, OCZ negative polarity | 0-3         | 0       | -    | U16       | At stop           | Upon re-power-on |

| Index | Parameter | Name          | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 31h   | C00.30    | Ordinary user | -       | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |
| 32h   | C00.31    | Super user    | -       | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |

## NOTICE

- For details about parameters above, refer to section 8.24.1 "Group C00".

## 8.2 Basic Gain Parameters (2001h/C01)

| Index | Parameter | Name                                         | Options                                                                                                         | Value Range | Default | Unit     | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------|---------|----------|-----------|-------------------|----------------|
| 01h   | C01.00    | 1st position loop gain                       | -                                                                                                               | 0-20000     | 400     | 0.1rad/s | U16       | During operation  | Immediately    |
| 02h   | C01.01    | 1st speed loop gain                          | -                                                                                                               | 1-20000     | 250     | 0.1Hz    | U16       | During operation  | Immediately    |
| 03h   | C01.02    | 1st speed loop integral time parameter       | -                                                                                                               | 1-51200     | 3184    | 0.01ms   | U16       | During operation  | Immediately    |
| 04h   | C01.03    | 1st torque reference filter cutoff frequency | -                                                                                                               | 5-16000     | 200     | Hz       | U16       | During operation  | Immediately    |
| 09h   | C01.08    | 2nd position loop gain                       | -                                                                                                               | 0-20000     | 560     | 0.1rad/s | U16       | During operation  | Immediately    |
| 0Ah   | C01.09    | 2nd speed loop gain                          | -                                                                                                               | 1-20000     | 350     | 0.1Hz    | U16       | During operation  | Immediately    |
| 08h   | C01.0A    | 2nd speed loop integral time parameter       | -                                                                                                               | 1-51200     | 2274    | 0.01ms   | U16       | During operation  | Immediately    |
| 0Ch   | C01.0B    | 2nd torque reference filter cutoff frequency | -                                                                                                               | 5-16000     | 280     | Hz       | U16       | During operation  | Immediately    |
| 11h   | C01.10    | Speed feedback filter                        | 0: Internal setting<br>1: Low-pass filter<br>2: Overlapping average filter<br>3: Speed observer<br>4: No filter | 0-4         | 0       | -        | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                                          | Options                                                                                                                                       | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------|-----------|-------------------|----------------|
| 12h   | C01.11    | Cutoff frequency of speed feedback low-pass filter            | -                                                                                                                                             | 10-16000    | 8000    | Hz    | U16       | During operation  | Immediately    |
| 13h   | C01.12    | Speed feedback overlapping average filter time constant       | 0: No filter<br>1: 2 times filter<br>2: 4 times filter<br>3: 8 times filter<br>4: 16 times filter<br>5: 32 times filter<br>6: 64 times filter | 0-6         | 0       | -     | U16       | During operation  | Immediately    |
| 14h   | C01.13    | Speed feedforward source                                      | 0: No feedforward<br>1: Internal reference<br>2: Model tracking<br>3: AI1<br>4: AI2                                                           | 0-4         | 0       | -     | U16       | During operation  | Immediately    |
| 15h   | C01.14    | Speed feedforward percentage                                  | -                                                                                                                                             | 0-2000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 16h   | C01.15    | Speed feedforward filter cutoff frequency                     | -                                                                                                                                             | 5-16000     | 318     | Hz    | U16       | During operation  | Immediately    |
| 17h   | C01.16    | Torque feedforward source                                     | 0: No feedforward<br>1: Internal reference<br>2: Model tracking<br>3: AI1<br>4: AI2                                                           | 0-4         | 0       | -     | U16       | During operation  | Immediately    |
| 18h   | C01.17    | Torque feedforward percentage                                 | -                                                                                                                                             | 0-2000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 19h   | C01.18    | Torque feedforward filter cutoff frequency                    | -                                                                                                                                             | 5-16000     | 318     | Hz    | U16       | During operation  | Immediately    |
| 1Ch   | C01.1B    | PDFF control coefficient                                      | -                                                                                                                                             | 0-1000      | 1000    | 0.1%  | U16       | During operation  | Immediately    |
| 1Dh   | C01.1C    | Damping factor control coefficient                            | -                                                                                                                                             | 0-1000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 21h   | C01.20    | Position reference overlapping average filter time constant A | -                                                                                                                                             | 0-1280      | 0       | 0.1ms | U16       | At stop           | Immediately    |
| 22h   | C01.21    | Position reference overlapping average filter time constant B | -                                                                                                                                             | 0-1280      | 0       | 0.1ms | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                               | Options                                                                                                       | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------|---------|-------|-----------|-------------------|----------------|
| 23h   | C01.22    | Position reference low-pass filter time constant A | -                                                                                                             | 0-65535     | 0       | 0.1ms | U16       | At stop           | Immediately    |
| 24h   | C01.23    | Position reference low-pass filter time constant B | -                                                                                                             | 0-65535     | 0       | 0.1ms | U16       | At stop           | Immediately    |
| 25h   | C01.24    | 1st notch filter frequency of position reference   | -                                                                                                             | 0-2000      | 0       | 0.1Hz | U16       | At stop           | Immediately    |
| 26h   | C01.25    | 1st notch filter width of position reference       | -                                                                                                             | 0-1000      | 0       | 0.1%  | U16       | At stop           | Immediately    |
| 27h   | C01.26    | 1st notch filter depth of position reference       | -                                                                                                             | 10-1000     | 1000    | 0.1%  | U16       | At stop           | Immediately    |
| 28h   | C01.27    | 2nd notch filter frequency of position reference   | -                                                                                                             | 0-2000      | 0       | 0.1Hz | U16       | At stop           | Immediately    |
| 29h   | C01.28    | 2nd notch filter width of position reference       | -                                                                                                             | 0-1000      | 0       | 0.1%  | U16       | At stop           | Immediately    |
| 2Ah   | C01.29    | 2nd notch filter depth of position reference       | -                                                                                                             | 10-1000     | 1000    | 0.1%  | U16       | At stop           | Immediately    |
| 28h   | C01.2A    | Position reference pre-charge filter time constant | -                                                                                                             | 0-1280      | 0       | 0.1ms | U16       | At stop           | Immediately    |
| 31h   | C01.30    | Adaptive notch mode                                | 0: Disabled<br>1: 1st notch<br>2: 2nd notch<br>3: Notch parameter reset<br>4: Resonance frequency tested only | 0-4         | 0       | -     | U16       | During operation  | Immediately    |
| 32h   | C01.31    | Adaptive notch test times                          | -                                                                                                             | 0-65535     | 0       | Times | U16       | At stop           | Immediately    |
| 33h   | C01.32    | Adaptive notch test frequency                      | -                                                                                                             | 0-8000      | 0       | Hz    | U16       | Read only         | Immediately    |
| 34h   | C01.33    | Adaptive notch test amplitude                      | -                                                                                                             | 0-5000      | 0       | 0.1%  | U16       | Read only         | Immediately    |
| 35h   | C01.34    | Adaptive notch target torque                       | -                                                                                                             | 0-2000      | 0       | 0.1%  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                          | Options                                                                                                                                                                                                                                                                                                                                          | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------|-----------|-------------------|----------------|
| 36h   | C01.35    | Adaptive notch test torque threshold          | -                                                                                                                                                                                                                                                                                                                                                | 0-2000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 39h   | C01.38    | Gain switchover mode                          | 0: Fixed to the 1st gain set<br>1: DI switchover<br>2: DI P-PI switchover<br>3: Torque reference<br>4: Speed reference<br>5: Speed feedback<br>6: Speed reference change rate<br>7: Position deviation<br>8: Position reference<br>9: Positioning completed signal<br>10: Position reference and speed feedback<br>11: Speed reference threshold | 0-11        | 0       | -     | U16       | At stop           | Immediately    |
| 3Ah   | C01.39    | Gain switchover time                          | -                                                                                                                                                                                                                                                                                                                                                | 10-10000    | 50      | 0.1ms | U16       | During operation  | Immediately    |
| 3Bh   | C01.3A    | Gain switchover threshold                     | -                                                                                                                                                                                                                                                                                                                                                | 0-65535     | 10      | -     | U16       | During operation  | Immediately    |
| 3Ch   | C01.3B    | Gain switchover loop width                    | -                                                                                                                                                                                                                                                                                                                                                | 0-65535     | 10      | -     | U16       | During operation  | Immediately    |
| 3Dh   | C01.3C    | Gain switchover position lock setting         | -                                                                                                                                                                                                                                                                                                                                                | 0-65535     | 0       | -     | U16       | During operation  | Immediately    |
| 3Eh   | C01.3D    | Gain switchover position gain conversion time | -                                                                                                                                                                                                                                                                                                                                                | 10-10000    | 50      | 0.1ms | U16       | During operation  | Immediately    |
| 41h   | C01.40    | Frequency of the 1st notch                    | -                                                                                                                                                                                                                                                                                                                                                | 10-8000     | 8000    | Hz    | U16       | During operation  | Immediately    |
| 42h   | C01.41    | Width level of the 1st notch                  | -                                                                                                                                                                                                                                                                                                                                                | 0-4000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 43h   | C01.42    | Depth level of the 1st notch                  | -                                                                                                                                                                                                                                                                                                                                                | 10-1000     | 1000    | 0.1%  | U16       | During operation  | Immediately    |
| 44h   | C01.43    | Frequency of the 2nd notch                    | -                                                                                                                                                                                                                                                                                                                                                | 10-8000     | 8000    | Hz    | U16       | During operation  | Immediately    |
| 45h   | C01.44    | Width level of the 2nd notch                  | -                                                                                                                                                                                                                                                                                                                                                | 0-4000      | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 46h   | C01.45    | Depth level of the 2nd notch                  | -                                                                                                                                                                                                                                                                                                                                                | 10-1000     | 1000    | 0.1%  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                         | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 47h   | C01.46    | Frequency of the 3rd notch   | -       | 10-8000     | 8000    | Hz   | U16       | During operation  | Immediately    |
| 48h   | C01.47    | Width level of the 3rd notch | -       | 0-4000      | 0       | 0.1% | U16       | During operation  | Immediately    |
| 49h   | C01.48    | Depth level of the 3rd notch | -       | 10-1000     | 1000    | 0.1% | U16       | During operation  | Immediately    |
| 4Ah   | C01.49    | Frequency of the 4th notch   | -       | 10-8000     | 8000    | Hz   | U16       | During operation  | Immediately    |
| 4Bh   | C01.4A    | Width level of the 4th notch | -       | 0-4000      | 0       | 0.1% | U16       | During operation  | Immediately    |
| 4Ch   | C01.4B    | Depth level of the 4th notch | -       | 10-1000     | 1000    | 0.1% | U16       | During operation  | Immediately    |
| 4Dh   | C01.4C    | Frequency of the 5th notch   | -       | 10-8000     | 8000    | Hz   | U16       | During operation  | Immediately    |
| 4Eh   | C01.4D    | Width level of the 5th notch | -       | 0-4000      | 0       | 0.1% | U16       | During operation  | Immediately    |
| 4Fh   | C01.4E    | Depth level of the 5th notch | -       | 10-1000     | 1000    | 0.1% | U16       | During operation  | Immediately    |

## NOTICE

- For details about parameters above, refer to section 8.24.2 "Group C01".

## 8.3 Advanced Gain Parameters (2002h/C02)

| Index | Parameter | Name                                          | Options                                      | Value Range | Default | Unit     | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------------------------------|----------------------------------------------|-------------|---------|----------|-----------|-------------------|----------------|
| 01h   | C02.00    | Model tracking control                        | 0: Disabled<br>1: Single mass model tracking | 0-1         | 0       | -        | U16       | At stop           | Immediately    |
| 02h   | C02.01    | Model tracking control gain                   | -                                            | 10-20000    | 500     | 0.1rad/s | U16       | During operation  | Immediately    |
| 03h   | C02.02    | Model tracking inertia correction coefficient | -                                            | 10-8000     | 1000    | 0.1%     | U16       | During operation  | Immediately    |



| Index | Parameter | Name                                                         | Options                                                                                                                                                      | Value Range          | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------|--------|-----------|-------------------|----------------|
| 09h   | C02.08    | Backlash compensation mode                                   | 0: No compensation<br>1: Compensation in forward direction<br>2: Compensation in reverse direction<br>3: Compensation in both forward and reverse directions | 0-3                  | 0       | -      | U16       | At stop           | Immediately    |
| 0Ah   | C02.09    | Number of backlash compensation pulses                       | -                                                                                                                                                            | $-2^{31}-(2^{31}-1)$ | 0       | P      | I32       | At stop           | Immediately    |
| 0Ch   | C02.0B    | Backlash compensation filter time constant                   | -                                                                                                                                                            | 0-65535              | 0       | 0.01ms | U16       | During operation  | Immediately    |
| 31h   | C02.30    | Speed observer gain                                          | -                                                                                                                                                            | 0-40000              | 0       | 0.1Hz  | U16       | During operation  | Immediately    |
| 32h   | C02.31    | Speed observer inertia correction                            | -                                                                                                                                                            | 10-8000              | 1000    | 0.1%   | U16       | During operation  | Immediately    |
| 33h   | C02.32    | Speed observer speed feedback cutoff frequency               | -                                                                                                                                                            | 0-16000              | 0       | Hz     | U16       | During operation  | Immediately    |
| 39h   | C02.38    | Frequency for vibration suppression 1                        | -                                                                                                                                                            | 10-20000             | 1000    | 0.1Hz  | U16       | During operation  | Immediately    |
| 3Ah   | C02.39    | Inertia correction for vibration suppression 1               | -                                                                                                                                                            | 10-8000              | 1000    | 0.1%   | U16       | During operation  | Immediately    |
| 3Bh   | C02.3A    | Low-pass filter correction for vibration suppression 1       | -                                                                                                                                                            | -9999-9999           | 0       | 0.1Hz  | I16       | During operation  | Immediately    |
| 3Ch   | C02.3B    | Correction of high-pass filter 1 for vibration suppression 1 | -                                                                                                                                                            | -9999-9999           | 0       | 0.1Hz  | I16       | During operation  | Immediately    |
| 3Dh   | C02.3C    | Frequency of high-pass filter 2 for vibration suppression 1  | -                                                                                                                                                            | 10-50000             | 20000   | 0.1Hz  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                                         | Options | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------------------------------------|---------|-------------|---------|-------|-----------|-------------------|----------------|
| 3Eh   | C02.3D    | Ratio of compensation 1 for vibration suppression 1          | -       | 0-20000     | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 3Fh   | C02.3E    | Ratio of compensation 2 for vibration suppression 1          | -       | 0-20000     | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 41h   | C02.40    | Frequency for vibration suppression 2                        | -       | 10-20000    | 1000    | 0.1Hz | U16       | During operation  | Immediately    |
| 42h   | C02.41    | Inertia correction for vibration suppression 2               | -       | 10-8000     | 1000    | 0.1%  | U16       | During operation  | Immediately    |
| 43h   | C02.42    | Low-pass filter correction for vibration suppression 2       | -       | -9999-9999  | 0       | 0.1Hz | I16       | During operation  | Immediately    |
| 44h   | C02.43    | Correction of high-pass filter 1 for vibration suppression 2 | -       | -9999-9999  | 0       | 0.1Hz | I16       | During operation  | Immediately    |
| 45h   | C02.44    | Frequency of high-pass filter 2 for vibration suppression 2  | -       | 10-50000    | 20000   | 0.1Hz | U16       | During operation  | Immediately    |
| 46h   | C02.45    | Ratio of compensation 1 for vibration suppression 2          | -       | 0-20000     | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 47h   | C02.46    | Ratio of compensation 2 for vibration suppression 2          | -       | 0-20000     | 0       | 0.1%  | U16       | During operation  | Immediately    |
| 49h   | C02.48    | Frequency for vibration suppression 3                        | -       | 10-20000    | 1000    | 0.1Hz | U16       | During operation  | Immediately    |
| 4Ah   | C02.49    | Inertia correction for vibration suppression 3               | -       | 10-8000     | 1000    | 0.1%  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                                         | Options | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------------------------------------|---------|-------------|---------|--------|-----------|-------------------|----------------|
| 4Bh   | C02.4A    | Low-pass filter correction for vibration suppression 3       | -       | -9999-9999  | 0       | 0.1Hz  | I16       | During operation  | Immediately    |
| 4Ch   | C02.4B    | Correction of high-pass filter 1 for vibration suppression 3 | -       | -9999-9999  | 0       | 0.1Hz  | I16       | During operation  | Immediately    |
| 4Dh   | C02.4C    | Frequency of high-pass filter 2 for vibration suppression 3  | -       | 10-50000    | 20000   | 0.1Hz  | U16       | During operation  | Immediately    |
| 4Eh   | C02.4D    | Ratio of compensation 1 for vibration suppression 3          | -       | 0-20000     | 0       | 0.1%   | U16       | During operation  | Immediately    |
| 4Fh   | C02.4E    | Ratio of compensation 2 for vibration suppression 3          | -       | 0-20000     | 0       | 0.1%   | U16       | During operation  | Immediately    |
| 61h   | C02.60    | Disturbance observer gain                                    | -       | 0-40000     | 0       | 0.1Hz  | U16       | During operation  | Immediately    |
| 62h   | C02.61    | Disturbance observer inertia correction coefficient          | -       | 1-10000     | 1000    | 0.1%   | U16       | During operation  | Immediately    |
| 63h   | C02.62    | Disturbance observer low-pass cutoff frequency               | -       | 0-16000     | 0       | Hz     | U16       | During operation  | Immediately    |
| 64h   | C02.63    | Disturbance observer compensation torque percentage          | -       | 0-2000      | 0       | 0.1%   | U16       | During operation  | Immediately    |
| 69h   | C02.68    | Friction compensation switch and relevant setting            | -       | 0-65535     | 0       | -      | U16       | During operation  | Immediately    |
| 6Ah   | C02.69    | Friction compensation speed threshold                        | -       | 0-5000      | 20      | 0.1rpm | U16       | During operation  | Immediately    |
| 68h   | C02.6A    | Static friction compensation                                 | -       | 0-2000      | 0       | 0.1%   | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                              | Options | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------------------|---------|-------------|---------|--------|-----------|-------------------|----------------|
| 6Ch   | C02.6B    | Forward friction compensation of coulomb friction | -       | 0-2000      | 0       | 0.1%   | U16       | During operation  | Immediately    |
| 6Dh   | C02.6C    | Reverse friction compensation of coulomb friction | -       | -2000-0     | 0       | 0.1%   | I16       | During operation  | Immediately    |
| 6Eh   | C02.6D    | Viscous friction torque for rated speed           | -       | 0-2000      | 0       | 0.1%   | U16       | During operation  | Immediately    |
| 6Fh   | C02.6E    | Friction compensation filter time                 | -       | 0-65535     | 0       | 0.01ms | U16       | During operation  | Immediately    |
| 70h   | C02.6F    | Friction compensation threshold for zero speed    | -       | 0-1000      | 10      | 0.1rpm | U16       | During operation  | Immediately    |

## 8.4 Instruction Parameters (2003h/C03)

| Index | Parameter | Name                                         | Options                                                               | Value Range    | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------------|-----------------------------------------------------------------------|----------------|---------|---------------------|-----------|-------------------|----------------|
| 01h   | C03.00    | Position reference selection                 | 0: Pulse reference<br>1: Internal position planning<br>2: Step amount | 0-2            | 0       | -                   | U16       | At stop           | Immediately    |
| 03h   | C03.02    | Numerator of group 1 electronic gear ratio   | -                                                                     | $1-(2^{32}-1)$ | 131072  | -                   | U32       | During operation  | Immediately    |
| 05h   | C03.04    | Denominator of group 1 electronic gear ratio | -                                                                     | $1-(2^{32}-1)$ | 10000   | -                   | U32       | During operation  | Immediately    |
| 07h   | C03.06    | Numerator of group 2 electronic gear ratio   | -                                                                     | $1-(2^{32}-1)$ | 131072  | -                   | U32       | During operation  | Immediately    |
| 09h   | C03.08    | Denominator of group 2 electronic gear ratio | -                                                                     | $1-(2^{32}-1)$ | 10000   | -                   | U32       | During operation  | Immediately    |
| 0Ch   | C03.0B    | Gear ratio switchover mode                   | 0: Switch after positioning is completed<br>1: Switch immediately     | 0-1            | 0       | -                   | U16       | During operation  | Immediately    |
| 0Dh   | C03.0C    | Step amount reference value                  | -                                                                     | -9999-9999     | 50      | Unit in application | I16       | During operation  | Immediately    |

| Index | Parameter | Name                          | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Value Range | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|---------------------|-----------|-------------------|----------------|
| 0Eh   | C03.0D    | Step amount speed reference   | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0-1500      | 1000    | rpm                 | U16       | During operation  | Immediately    |
| 11h   | C03.10    | Position reached criteria     | 0: DO output when the loop position deviation is less than the threshold<br>1: DO output when the loop position deviation is less than the threshold and the filtered position reference is 0<br>2. DO output when the loop position deviation is less than the threshold and the filtered position reference is 0 and the output is kept active for at least the time set by C03.14<br>3: DO output when the actual position deviation is less than the threshold<br>4: DO output when the actual position deviation is less than the decision threshold and the filtered position reference is 0<br>5. DO output when the actual position deviation is less than the threshold, and the filtered position reference is 0 and the output is kept active for at least the time set by C03.14 | 0-5         | 1       | -                   | U16       | At stop           | Immediately    |
| 12h   | C03.11    | Threshold of positioning near | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0-65535     | 625     | Unit in application | U16       | During operation  | Immediately    |
| 13h   | C03.12    | Position reached threshold    | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0-65535     | 7       | Unit in application | U16       | During operation  | Immediately    |
| 14h   | C03.13    | Position reached filter time  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0-65535     | 0       | ms                  | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                  | Options                                                                                                                                                 | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 15h   | C03.14    | Position reached holding time         | -                                                                                                                                                       | 0-65535     | 0       | ms   | U16       | At stop           | Immediately    |
| 17h   | C03.16    | Position deviation clearing selection | 0: Inactive<br>1: DI trigger                                                                                                                            | 0-1         | 0       | -    | U16       | At stop           | Immediately    |
| 21h   | C03.20    | Speed reference selection             | 0: Internal digital reference<br>1: AI1<br>2: AI2<br>3: Internal speed planning<br>4: AI1 → internal speed planning<br>5: AI2 → internal speed planning | 0-5         | 0       | -    | U16       | At stop           | Immediately    |
| 22h   | C03.21    | Speed reference                       | -                                                                                                                                                       | -8000-8000  | 100     | rpm  | I16       | During operation  | Immediately    |
| 23h   | C03.22    | Acceleration rate                     | -                                                                                                                                                       | 0-65535     | 10      | ms   | U32       | During operation  | Immediately    |
| 25h   | C03.24    | Deceleration rate                     | -                                                                                                                                                       | 0-65535     | 10      | ms   | U32       | During operation  | Immediately    |
| 27h   | C03.26    | Speed limit selection                 | 0: Internal limit<br>1: External limit<br>2: AI1<br>3: AI2                                                                                              | 0-3         | 0       | -    | U16       | At stop           | Immediately    |
| 28h   | C03.27    | Internal positive speed limit         | -                                                                                                                                                       | 0-8000      | 6000    | rpm  | U16       | During operation  | Immediately    |
| 29h   | C03.28    | Internal negative speed limit         | -                                                                                                                                                       | 0-8000      | 6000    | rpm  | U16       | During operation  | Immediately    |
| 2Ah   | C03.29    | External forward speed limit          | -                                                                                                                                                       | 0-8000      | 6000    | rpm  | U16       | During operation  | Immediately    |
| 2Bh   | C03.2A    | External negative speed limit         | -                                                                                                                                                       | 0-8000      | 6000    | rpm  | U16       | During operation  | Immediately    |
| 2Ch   | C03.2B    | Speed reach threshold                 | -                                                                                                                                                       | 0-8000      | 1000    | rpm  | U16       | During operation  | Immediately    |
| 2Dh   | C03.2C    | Speed synchronization threshold       | -                                                                                                                                                       | 0-1000      | 10      | rpm  | U16       | During operation  | Immediately    |
| 2Eh   | C03.2D    | Speed rotation threshold              | -                                                                                                                                                       | 0-1000      | 20      | rpm  | U16       | During operation  | Immediately    |
| 2Fh   | C03.2E    | Zero speed output threshold           | -                                                                                                                                                       | 5-1000      | 10      | rpm  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                                          | Options                                                    | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------------------------------|------------------------------------------------------------|-------------|---------|-------|-----------|-------------------|----------------|
| 30h   | C03.2F    | Zero clamp threshold                                          | -                                                          | 0-1000      | 10      | rpm   | U16       | During operation  | Immediately    |
| 31h   | C03.30    | Forward DI jog speed                                          | -                                                          | 0-8000      | 100     | rpm   | U16       | During operation  | Immediately    |
| 32h   | C03.31    | Reverse DI jog speed                                          | -                                                          | 0-8000      | 100     | rpm   | U16       | During operation  | Immediately    |
| 33h   | C03.32    | Zero-speed output judgment hysteresis                         | -                                                          | 0-200       | 10      | rpm   | U16       | During operation  | Immediately    |
| 34h   | C03.33    | Speed DO filter setting                                       | -                                                          | 0-1000      | 100     | 0.1ms | U16       | During operation  | Immediately    |
| 35h   | C03.34    | Speed S-curve enable                                          | 0: Disabled<br>1: Enabled                                  | 0-1         | 0       | -     | U16       | At stop           | Immediately    |
| 36h   | C03.35    | Increasing acceleration of speed S-curve acceleration segment | -                                                          | 0-1000      | 500     | 0.1%  | U16       | At stop           | Immediately    |
| 37h   | C03.36    | Increasing deceleration of speed S-curve acceleration segment | -                                                          | 0-1000      | 500     | 0.1%  | U16       | At stop           | Immediately    |
| 38h   | C03.37    | Decreasing acceleration of speed S-curve deceleration segment | -                                                          | 0-1000      | 500     | 0.1%  | U16       | At stop           | Immediately    |
| 39h   | C03.38    | Decreasing deceleration of speed S-curve deceleration segment | -                                                          | 0-1000      | 500     | 0.1%  | U16       | At stop           | Immediately    |
| 41h   | C03.40    | Torque reference selection                                    | 0: Internal digital setting<br>1: AI1<br>2: AI2            | 0-2         | 0       | -     | U16       | At stop           | Immediately    |
| 42h   | C03.41    | Torque reference                                              | -                                                          | -4000-4000  | 0       | 0.1%  | I16       | During operation  | Immediately    |
| 43h   | C03.42    | Torque limit selection                                        | 0: Internal limit<br>1: External limit<br>2: AI1<br>3: AI2 | 0-3         | 0       | -     | U16       | At stop           | Immediately    |
| 44h   | C03.43    | Internal positive torque limit                                | -                                                          | 0-4000      | 3000    | 0.1%  | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 45h   | C03.44    | Internal negative torque limit      | -       | 0-4000      | 3000    | 0.1% | U16       | During operation  | Immediately    |
| 46h   | C03.45    | External forward torque limit       | -       | 0-4000      | 3000    | 0.1% | U16       | During operation  | Immediately    |
| 47h   | C03.46    | External reverse torque limit       | -       | 0-4000      | 3000    | 0.1% | U16       | During operation  | Immediately    |
| 48h   | C03.47    | Positive speed limit in torque mode | -       | 0-8000      | 3000    | rpm  | U16       | During operation  | Immediately    |
| 49h   | C03.48    | Negative speed limit in torque mode | -       | 0-8000      | 3000    | rpm  | U16       | During operation  | Immediately    |
| 4Ah   | C03.49    | Reference value for torque reach    | -       | 0-4000      | 0       | 0.1% | U16       | During operation  | Immediately    |
| 4Bh   | C03.4A    | Valid value for torque reached      | -       | 0-4000      | 200     | 0.1% | U16       | During operation  | Immediately    |
| 4Ch   | C03.4B    | Invalid value for torque reached    | -       | 0-4000      | 100     | 0.1% | U16       | During operation  | Immediately    |

## NOTICE

- For details about parameters above, refer to section 8.24.3 "Group C03".



## 8.5 I/O Parameters (2004h/C04)

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | C04.00    | DI1 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 6       | -    | U16       | At stop           | Immediately    |
| 02h   | C04.01    | DI1 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 03h   | C04.02    | DI1 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 05h   | C04.04    | DI2 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 7       | -      | U16       | At stop           | Immediately    |
| 06h   | C04.05    | DI2 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 07h   | C04.06    | DI2 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 09h   | C04.08    | DI3 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 24      | -      | U16       | At stop           | Immediately    |
| 0Ah   | C04.09    | DI3 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 0Bh   | C04.0A    | DI3 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 0Dh   | C04.0C    | DI4 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 2       | -      | U16       | At stop           | Immediately    |
| 0Eh   | C04.0D    | DI4 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 0Fh   | C04.0E    | DI4 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 11h   | C04.10    | DI5 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 1       | -      | U16       | At stop           | Immediately    |
| 12h   | C04.11    | DI5 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 13h   | C04.12    | DI5 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 15h   | C04.14    | DI6 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 0       | -      | U16       | At stop           | Immediately    |
| 16h   | C04.15    | DI6 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 17h   | C04.16    | DI6 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 19h   | C04.18    | DI7 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 33      | -      | U16       | At stop           | Immediately    |
| 1Ah   | C04.19    | DI7 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 18h   | C04.1A    | DI7 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 1Dh   | C04.1C    | DI8 function selection | 0: No definition<br>1: S-ON<br>2: Fault reset<br>3: Gain switchover<br>4: Emergency stop<br>5: Home switch<br>6: Forward overtravel<br>7: Reverse overtravel<br>8: Mode 1<br>9: Mode 2<br>10: Positive jog<br>11: Negative jog<br>12: Segment selection 1<br>13: Segment selection 2<br>14: Segment selection 3<br>15: Segment selection 4<br>16: Position reference direction<br>17: Velocity reference direction<br>18: Torque reference direction<br>19: Position planning trigger<br>20: Position planning pause<br>21: Position capture inhibit<br>22: Position capture release<br>23: Position deviation clear<br>24: Position reference inhibit<br>25: Referencing start enable<br>26: Velocity reference mode<br>27: Zero clamp<br>28: External velocity limit<br>29: External torque limit<br>32: Electronic gear ratio switchover<br>33: Position capture enable<br>34: Position step amount enable<br>35: Hand wheel enable<br>36: Hand wheel ratio 1<br>37: Hand wheel ratio 2 | 0-37        | 5       | -      | U16       | At stop           | Immediately    |
| 1Eh   | C04.1D    | DI8 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0-1         | 0       | -      | U16       | During operation  | Immediately    |



| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 1Fh   | C04.1E    | DI8 filter time        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0-65535     | 150     | 0.01ms | U16       | During operation  | Immediately    |
| 31h   | C04.30    | DO1 function selection | 0: No definition<br>1: Servo ready<br>2: Motor rotation<br>3: Brake<br>4: Fault<br>5: Alarm<br>6: Servo operation<br>7: Position reach<br>8: Positioning near<br>9: Referencing completion<br>10: Position capture completion<br>11: Position comparison output<br>12: Zero velocity signal<br>13: Velocity limit<br>14: Velocity reach<br>15: Velocity synchronization<br>16: Torque limit<br>17: Torque reached<br>18: EMD output<br>19: Angle auto-tuning completion | 0-20        | 1       | -      | U16       | At stop           | Immediately    |
| 32h   | C04.31    | DO1 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-1         | 0       | -      | U16       | During operation  | Immediately    |
| 33h   | C04.32    | DO2 function selection | 0: No definition<br>1: Servo ready<br>2: Motor rotation<br>3: Brake<br>4: Fault<br>5: Alarm<br>6: Servo operation<br>7: Position reach<br>8: Positioning near<br>9: Referencing completion<br>10: Position capture completion<br>11: Position comparison output<br>12: Zero velocity signal<br>13: Velocity limit<br>14: Velocity reach<br>15: Velocity synchronization<br>16: Torque limit<br>17: Torque reached<br>18: EMD output<br>19: Angle auto-tuning completion | 0-20        | 7       | -      | U16       | At stop           | Immediately    |

| Index | Parameter | Name                   | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 34h   | C04.33    | DO2 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-1         | 0       | -    | U16       | During operation  | Immediately    |
| 35h   | C04.34    | DO3 function selection | 0: No definition<br>1: Servo ready<br>2: Motor rotation<br>3: Brake<br>4: Fault<br>5: Alarm<br>6: Servo operation<br>7: Position reach<br>8: Positioning near<br>9: Referencing completion<br>10: Position capture completion<br>11: Position comparison output<br>12: Zero velocity signal<br>13: Velocity limit<br>14: Velocity reach<br>15: Velocity synchronization<br>16: Torque limit<br>17: Torque reached<br>18: EMD output<br>19: Angle auto-tuning completion | 0-20        | 3       | -    | U16       | At stop           | Immediately    |
| 36h   | C04.35    | DO3 logic selection    | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-1         | 0       | -    | U16       | During operation  | Immediately    |
| 37h   | C04.36    | DO4 function selection | 0: No definition<br>1: Servo ready<br>2: Motor rotation<br>3: Brake<br>4: Fault<br>5: Alarm<br>6: Servo operation<br>7: Position reach<br>8: Positioning near<br>9: Referencing completion<br>10: Position capture completion<br>11: Position comparison output<br>12: Zero velocity signal<br>13: Velocity limit<br>14: Velocity reach<br>15: Velocity synchronization<br>16: Torque limit<br>17: Torque reached<br>18: EMD output<br>19: Angle auto-tuning completion | 0-20        | 4       | -    | U16       | At stop           | Immediately    |

| Index | Parameter | Name                      | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 38h   | C04.37    | DO4 logic selection       | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-1         | 0       | -      | U16       | During operation  | Immediately    |
| 39h   | C04.38    | DO5 function selection    | 0: No definition<br>1: Servo ready<br>2: Motor rotation<br>3: Brake<br>4: Fault<br>5: Alarm<br>6: Servo operation<br>7: Position reach<br>8: Positioning near<br>9: Referencing completion<br>10: Position capture completion<br>11: Position comparison output<br>12: Zero velocity signal<br>13: Velocity limit<br>14: Velocity reach<br>15: Velocity synchronization<br>16: Torque limit<br>17: Torque reached<br>18: EMD output<br>19: Angle auto-tuning completion | 0-20        | 9       | -      | U16       | At stop           | Immediately    |
| 3Ah   | C04.39    | DO5 logic selection       | 0: NO<br>1: NC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0-1         | 0       | -      | U16       | During operation  | Immediately    |
| 41h   | C04.40    | AI1 offset                | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -5000-5000  | 0       | mV     | I16       | During operation  | Immediately    |
| 42h   | C04.41    | AI1 dead zone             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0-50000     | 0       | 0.1mV  | U16       | During operation  | Immediately    |
| 43h   | C04.42    | AI1 zero drift            | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -5000-5000  | 0       | 0.1mV  | I16       | During operation  | Immediately    |
| 44h   | C04.43    | AI1 median filter enable  | 0: Inactive<br>1: Enabled                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0-1         | 1       | -      | U16       | During operation  | Immediately    |
| 45h   | C04.44    | AI1 low-pass filter time  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0-65535     | 50      | 0.01ms | U16       | During operation  | Immediately    |
| 46h   | C04.45    | AI function configuration | Bit00: Current-type AI disconnection detection switch<br>0: Disabled<br>1: Open                                                                                                                                                                                                                                                                                                                                                                                         | 0-65535     | 0       | -      | U16       | At stop           | Immediately    |
| 4Ah   | C04.49    | AI2 median filter enable  | 0: Inactive<br>1: Active                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0-1         | 1       | -      | U16       | During operation  | Immediately    |

| Index | Parameter | Name                                       | Options                                                                                                                                                                                                                                                                                                                                                                                        | Value Range | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|--------|-----------|-------------------|----------------|
| 48h   | C04.4A    | AI2 low-pass filter time                   | -                                                                                                                                                                                                                                                                                                                                                                                              | 0-65535     | 50      | 0.01ms | U16       | During operation  | Immediately    |
| 4Ch   | C04.4B    | Speed value corresponding to analog 10 V   | -                                                                                                                                                                                                                                                                                                                                                                                              | 0-20000     | 3000    | rpm    | U16       | During operation  | Immediately    |
| 4Dh   | C04.4C    | Torque value corresponding to analog 10 V  | -                                                                                                                                                                                                                                                                                                                                                                                              | 0-8000      | 1000    | 0.1%   | U16       | During operation  | Immediately    |
| 4Eh   | C04.4D    | Speed value corresponding to analog 20 mA  | -                                                                                                                                                                                                                                                                                                                                                                                              | 0-20000     | 3000    | rpm    | U16       | During operation  | Immediately    |
| 4Fh   | C04.4E    | Torque value corresponding to analog 20 mA | -                                                                                                                                                                                                                                                                                                                                                                                              | 0-8000      | 1000    | 0.1%   | U16       | During operation  | Immediately    |
| 51h   | C04.50    | AO1 mode selection                         | 0: Speed feedback<br>1: Speed reference<br>2: Torque reference<br>3: Position deviation (reference unit)<br>4: Position deviation (encoder unit)<br>5: Position reference speed<br>6: Positioning completed<br>7: Speed feedforward<br>8: AI1 sampling value<br>9: AI2 sampling value<br>10: Torque feedforward<br>11: Bus voltage<br>12: End of position reference output<br>16: Voltage test | 0-16        | 0       | -      | U16       | During operation  | Immediately    |
| 52h   | C04.51    | AO1 offset voltage                         | -                                                                                                                                                                                                                                                                                                                                                                                              | -9999-9999  | 0       | mV     | I16       | During operation  | Immediately    |
| 53h   | C04.52    | AO1 ratio                                  | -                                                                                                                                                                                                                                                                                                                                                                                              | -9999-9999  | 100     | 0.01   | I16       | During operation  | Immediately    |

## 8.6 Stop Mode (2005h/C05)

| Index | Parameter | Name                  | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | C05.00    | Stop mode at S-ON Off | -4: Ramp to stop (C05.08), keeping the DB state<br>-3: Stop at zero speed, keeping the DB state<br>-2: Ramp to stop (C05.0A), keeping the DB state<br>-1: DB stop, keeping the DB state<br>0: Coast to stop, keeping the de-energized state<br>1: Ramp to stop (C05.0A), keeping the de-energized state<br>2: Stop at zero speed, keeping the de-energized state                                                                                                                         | -4-2        | 0       | -    | I16       | At stop           | Immediately    |
| 02h   | C05.01    | Quick stop mode       | 0: Coast to stop, keeping the de-energized state<br>1: Ramp to stop (C05.0A), keeping the de-energized state<br>2: Ramp to stop (C05.08), keeping the de-energized state<br>3: Stop at emergency stop torque, keeping the de-energized state<br>4: Stop at zero speed, keeping the de-energized state<br>5: Ramp to stop (C05.0A), keeping the locked position<br>6: Ramp to stop (C05.08), keeping the locked position<br>7: Stop at emergency stop torque, keeping the locked position | 0-7         | 2       | -    | U16       | At stop           | Immediately    |

| Index | Parameter | Name                     | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 03h   | C05.02    | Stop mode at overtravel  | 0: Coast to stop, keeping de-energized status<br>1: Stop at zero speed, keeping position lock status<br>2: Stop at zero speed, keeping de-energized status<br>3: Ramp to stop as defined by C05.08, keeping de-energized status<br>4: Ramp to stop as defined by C05.08, keeping position lock status<br>5: Dynamic braking stop, keeping de-energized status<br>6: Dynamic braking stop, keeping dynamic braking status<br>7: Not responding to overtravel | 0-7         | 1       | -    | U16       | At stop           | Immediately    |
| 04h   | C05.03    | Stop mode at No. 1 fault | 0: Coast to stop, keeping de-energized status<br>1: Dynamic braking stop, keeping de-energized status<br>2: Dynamic braking stop, keeping dynamic braking status                                                                                                                                                                                                                                                                                            | 0-2         | 2       | -    | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                                | Options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 05h   | C05.04    | Stop mode at No. 2 fault                            | -5: Stop at zero speed, keeping the DB state<br>-4: Stop at emergency stop torque, keeping the DB state<br>-3: Ramp to stop (C05.08), keeping the DB state<br>-2: Ramp to stop (C05.0A), keeping the DB state<br>-1: DB stop, keeping the DB state<br>0: Coast to stop, keeping the de-energized state<br>1: Ramp to stop (C05.0A), keeping the de-energized state<br>2: Ramp to stop (C05.08), keeping the de-energized state<br>3: Stop at emergency stop torque, keeping the de-energized state | -5-3        | 1       | -    | I16       | At stop           | Immediately    |
| 08h   | C05.07    | Stop speed threshold                                | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5-1000      | 10      | rpm  | U16       | During operation  | Immediately    |
| 09h   | C05.08    | Deceleration time for ramp to quick stop            | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0-3600000   | 0       | ms   | U32       | During operation  | Immediately    |
| 08h   | C05.0A    | Deceleration time for ramp to slow stop             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0-3600000   | 100     | ms   | U32       | During operation  | Immediately    |
| 0Dh   | C05.0C    | Limit for stop at emergency-stop torque             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5-3000      | 1000    | 0.1% | U16       | During operation  | Immediately    |
| 11h   | C05.10    | Delay from brake close to motor de-energized        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0-65535     | 100     | ms   | U16       | During operation  | Immediately    |
| 12h   | C05.11    | Speed threshold at brake closing                    | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10-3000     | 30      | rpm  | U16       | During operation  | Immediately    |
| 13h   | C05.12    | Maximum waiting time with S-ON off at brake closing | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0-65535     | 100     | ms   | U16       | During operation  | Immediately    |
| 14h   | C05.13    | Delay from brake on to command received             | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0-65535     | 100     | ms   | U16       | At stop           | Immediately    |

| Index | Parameter | Name                         | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 15h   | C05.14    | Energizing delay of DB relay | -       | 0-65535     | 20      | ms   | U16       | At stop           | Immediately    |

## 8.7 Protection Parameters (2006h/C06)

| Index | Parameter | Name                                                | Options                                              | Value Range          | Default    | Unit                | Data Type | Modification Mode | Effective Time   |
|-------|-----------|-----------------------------------------------------|------------------------------------------------------|----------------------|------------|---------------------|-----------|-------------------|------------------|
| 01h   | C06.00    | Threshold of excessive local position deviation     | -                                                    | $0-(2^{32}-1)$       | 32767      | Unit in application | U32       | During operation  | Immediately      |
| 03h   | C06.02    | Threshold of excessive position reference frequency | -                                                    | 0-4000               | 4000       | kHz                 | U16       | At stop           | Immediately      |
| 04h   | C06.03    | Threshold of excessive speed                        | -                                                    | 0-9000               | 0          | rpm                 | U16       | During operation  | Immediately      |
| 05h   | C06.04    | Input phase loss detection                          | 0: Enabled<br>1: Disabled                            | 0-1                  | 0          | -                   | U16       | At stop           | Immediately      |
| 06h   | C06.05    | Retentive at power failure                          | 0: Non-retentive<br>1: Retentive                     | 0-1                  | 0          | -                   | U16       | At stop           | Immediately      |
| 07h   | C06.06    | STO function shielding                              | -                                                    | 0-1                  | 0          | -                   | U16       | At stop           | Immediately      |
| 08h   | C06.07    | Mechanical limit position                           | 0: Inactive<br>1: Enabled<br>2: Enabled after homing | 0-2                  | 0          | -                   | U16       | During operation  | Immediately      |
| 09h   | C06.08    | Mechanical PL                                       | -                                                    | $-2^{31}-(2^{31}-1)$ | $2^{31}-1$ | Unit in application | I32       | During operation  | Immediately      |
| 0Bh   | C06.0A    | Mechanical NL                                       | -                                                    | $-2^{31}-(2^{31}-1)$ | $-2^{31}$  | Unit in application | I32       | During operation  | Immediately      |
| 0Eh   | C06.0D    | Three-phase 220 V to two-phase alarm switch         | 0: Disabled<br>1: Enabled                            | 0-1                  | 0          | -                   | U16       | At stop           | Immediately      |
| 0Fh   | C06.0E    | STO handling method                                 | 0: Fault<br>1: State                                 | 0-1                  | 1          | -                   | U16       | At stop           | Upon re-power-on |
| 11h   | C06.10    | Drive overload protection threshold                 | -                                                    | 0-3500               | 1150       | 0.1%                | U16       | At stop           | Immediately      |
| 12h   | C06.11    | Motor overload protection threshold                 | -                                                    | 0-3500               | 1150       | 0.1%                | U16       | At stop           | Immediately      |



| Index | Parameter | Name                                                    | Options                   | Value Range | Default | Unit      | Data Type | Modification Mode | Effective Time   |
|-------|-----------|---------------------------------------------------------|---------------------------|-------------|---------|-----------|-----------|-------------------|------------------|
| 13h   | C06.12    | Motor locked-rotor detection                            | 0: Inactive<br>1: Enabled | 0-1         | 1       | -         | U16       | At stop           | Immediately      |
| 14h   | C06.13    | Motor locked-rotor detection time                       | -                         | 0-3000      | 200     | ms        | U16       | At stop           | Immediately      |
| 15h   | C06.14    | Motor locked-rotor detection speed                      | -                         | 0-1000      | 10      | rpm       | U16       | During operation  | Immediately      |
| 16h   | C06.15    | Output phase loss detection                             | 0: Inactive<br>1: Enabled | 0-1         | 0       | -         | U16       | At stop           | Immediately      |
| 17h   | C06.16    | PTC enabling                                            | 0: Inactive<br>1: Enabled | 0-1         | 0       | -         | U16       | At stop           | Immediately      |
| 18h   | C06.17    | Motor overload current coefficient                      | -                         | 0-200       | 100     | %         | U16       | At stop           | Immediately      |
| 19h   | C06.18    | IGBT over-temperature protection threshold              | -                         | 0-200       | 135     | °C        | U16       | At stop           | Upon re-power-on |
| 1Ah   | C06.19    | Discharge conduit over-temperature protection threshold | -                         | 0-200       | 115     | °C        | U16       | At stop           | Upon re-power-on |
| 18h   | C06.1A    | Encoder over-temperature protection threshold           | -                         | 0-200       | 80      | °C        | U16       | At stop           | Upon re-power-on |
| 1Ch   | C06.1B    | Encoder multi-turn overflow protection                  | 0: Inactive<br>1: Enabled | 0-1         | 1       | -         | U16       | At stop           | Immediately      |
| 1Dh   | C06.1C    | Drive high temperature warning threshold                | -                         | 0-1200      | 900     | 0.1°C     | U16       | During operation  | Immediately      |
| 1Eh   | C06.1D    | Output phase loss detection threshold                   | -                         | 0-2000      | 0       | 0.1 times | U16       | During operation  | Immediately      |
| 21h   | C06.20    | Protection from out of control                          | 0: Inactive<br>1: Enabled | 0-1         | 1       | -         | U16       | At stop           | Immediately      |
| 22h   | C06.21    | Torque threshold for protection from out of control     | -                         | 1-4000      | 2000    | 0.1%      | U16       | During operation  | Immediately      |

| Index | Parameter | Name                                                 | Options | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time   |
|-------|-----------|------------------------------------------------------|---------|-------------|---------|-------|-----------|-------------------|------------------|
| 23h   | C06.22    | Speed threshold for protection from out of control   | -       | 1-1000      | 50      | rpm   | U16       | During operation  | Immediately      |
| 24h   | C06.23    | Speed filter time for protection from out of control | -       | 1-1000      | 20      | 0.1ms | U16       | During operation  | Immediately      |
| 25h   | C06.24    | Detection time for protection from out of control    | -       | 10-1000     | 30      | ms    | U16       | During operation  | Immediately      |
| 26h   | C06.25    | Motor overtemperature protection enable              | -       | 0-1         | 0       | -     | U16       | During operation  | Immediately      |
| 27h   | C06.26    | Motor overtemperature protection threshold           | -       | 0-2000      | 1100    | 0.1°C | U16       | During operation  | Immediately      |
| 28h   | C06.27    | STO 24 V disconnection filter time                   | -       | 1-5         | 4       | ms    | U16       | At stop           | Upon re-power-on |
| 29h   | C06.28    | Servo OFF delay after STO triggered                  | -       | 0-25        | 20      | ms    | U16       | During operation  | Immediately      |
| 2Ah   | C06.29    | SizeCDE electromagnetic shorting                     | -       | 0-1         | 0       | -     | U16       | At stop           | Upon re-power-on |
| 2Ch   | C06.2B    | STO PWM buffer detection method                      | -       | 0-2         | 0       | -     | U16       | At stop           | Immediately      |

## NOTICE

- For details about parameters above, refer to section 8.24.4 "Group C06".

## 8.8 Auto-tuning Parameters (2007h/C07)

| Index | Parameter | Name                                        | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | C07.00    | Offline inertia auto-tuning mode setting    | -       | 0-785       | 769     | -    | U16       | At stop           | Immediately    |
| 02h   | C07.01    | Offline inertia auto-tuning speed reference | -       | 50-1000     | 500     | rpm  | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                                                                 | Options                                                                                               | Value Range  | Default | Unit   | Data Type | Modification Mode   | Effective Time       |
|-------|-----------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------|---------|--------|-----------|---------------------|----------------------|
| 03h   | C07.02    | Acceleration/<br>Deceleration time for<br>offline inertia auto-<br>tuning            | -                                                                                                     | 0-65535      | 100     | ms     | U16       | At stop             | Immediately          |
| 04h   | C07.03    | Offline inertia auto-<br>tuning target torque                                        | -                                                                                                     | 1-1500       | 150     | 0.1%   | U16       | At stop             | Immediately          |
| 05h   | C07.04    | Offline inertia auto-<br>tuning revolutions                                          | -                                                                                                     | 10-<br>65535 | 200     | 0.01r  | U16       | At stop             | Immediately          |
| 09h   | C07.08    | Online inertia auto-<br>tuning mode setting                                          | 0: Disabled<br>1: Slow inertia<br>change<br>2: Common inertia<br>change<br>3: Quick inertia<br>change | 0-3          | 0       | -      | U16       | During<br>operation | Immediately          |
| 0Ah   | C07.09    | Online inertia auto-<br>tuning parameter<br>setting                                  | -                                                                                                     | 0-65535      | 19      | -      | U16       | During<br>operation | Immediately          |
| 0Bh   | C07.0A    | Online inertia<br>auto-tuning speed<br>feedback over-lapping<br>filter time          | -                                                                                                     | 0-400        | 100     | 0.01ms | U16       | At stop             | Upon re-<br>power-on |
| 0Ch   | C07.0B    | Online inertia auto-<br>tuning torque<br>feedback over-lapping<br>filter time        | -                                                                                                     | 0-400        | 400     | 0.01ms | U16       | At stop             | Upon re-<br>power-on |
| 29h   | C07.28    | Low-speed area<br>threshold in friction<br>identification<br>reference               | -                                                                                                     | 10-200       | 100     | rpm    | U16       | At stop             | Immediately          |
| 2Ah   | C07.29    | Number of target<br>velocities in friction<br>identification<br>reference            | -                                                                                                     | 10-50        | 40      | -      | U16       | At stop             | Immediately          |
| 2Bh   | C07.2A    | Maximum speed in<br>friction identification<br>reference                             | -                                                                                                     | 100-500      | 500     | rpm    | U16       | At stop             | Immediately          |
| 2Ch   | C07.2B    | Number of velocities<br>in low-speed mode in<br>friction identification<br>reference | -                                                                                                     | 10-25        | 20      | -      | U16       | At stop             | Immediately          |

| Index | Parameter | Name                                                                  | Options | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------------------------------------------------------|---------|-------------|---------|-------|-----------|-------------------|----------------|
| 2Dh   | C07.2C    | Acceleration/Deceleration time in friction identification reference   | -       | 10-1000     | 120     | ms    | U16       | At stop           | Immediately    |
| 2Eh   | C07.2D    | Constant-velocity operation time in friction identification reference | -       | 10-1000     | 120     | ms    | U16       | At stop           | Immediately    |
| 2Fh   | C07.2E    | Stop time in friction identification reference                        | -       | 10-1000     | 120     | ms    | U16       | At stop           | Immediately    |
| 30h   | C07.2F    | Completion flag in friction identification reference                  | -       | 50-20000    | 200     | 0.01r | U16       | At stop           | Immediately    |

## 8.9 Communication Parameters (200Ah/C0A)

| Index | Parameter | Name                                   | Options                                                                                                                                                                       | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|------------------|
| 01h   | C0A.00    | Modbus communication station number    | -                                                                                                                                                                             | 1-255       | 1       | -    | U16       | At stop           | Immediately      |
| 02h   | C0A.01    | Modbus communication baud rate         | 1: 2400bps<br>2: 4800bps<br>3: 9600bps<br>4: 19200bps<br>5: 38400bps<br>6: 57600bps<br>7: 115200bps                                                                           | 1-7         | 7       | -    | U16       | At stop           | Upon re-power-on |
| 03h   | C0A.02    | Modbus communication format            | 0: No parity, 1 stop bit<br>1: Odd parity, 1 stop bit<br>2: Even parity, 1 stop bit<br>3: No parity, 2 stop bits<br>4: Odd parity, 2 stop bits<br>5: Even parity, 2 stop bits | 0-5         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 04h   | C0A.03    | Modbus communication response time     | -                                                                                                                                                                             | 1-1000      | 1       | ms   | U16       | At stop           | Immediately      |
| 06h   | C0A.05    | Modbus communication storage selection | 0: No storage<br>1: Storage                                                                                                                                                   | 0-1         | 1       | -    | U16       | At stop           | Immediately      |

| Index | Parameter | Name                                               | Options                                                                                                                                                                       | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|------------------|
| 07h   | COA.06    | Modbus data format                                 | 0: Low 16 bits before high 16 bits<br>1: High 16 bits before low 16 bits                                                                                                      | 0-1         | 0       | -    | U16       | At stop           | Immediately      |
| 09h   | COA.08    | Commissioning software communication station ID    | -                                                                                                                                                                             | 1-255       | 1       | -    | U16       | At stop           | Immediately      |
| 0Ah   | COA.09    | Commissioning software communication baud rate     | 0: 1200bps<br>1: 2400bps<br>2: 4800bps<br>3: 9600bps<br>4: 19200bps<br>5: 38400bps<br>6: 57600bps<br>7: 115200bps                                                             | 0-7         | 7       | -    | U16       | At stop           | Upon re-power-on |
| 0Bh   | COA.0A    | Commissioning software communication format        | 0: No parity, 1 stop bit<br>1: Odd parity, 1 stop bit<br>2: Even parity, 1 stop bit<br>3: No parity, 2 stop bits<br>4: Odd parity, 2 stop bits<br>5: Even parity, 2 stop bits | 0-5         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 0Ch   | COA.0B    | Commissioning software communication response time | -                                                                                                                                                                             | 1-1000      | 1       | ms   | U16       | At stop           | Immediately      |
| 0Eh   | COA.0D    | Commissioning software communication storage       | 0: No storage<br>1: Storage                                                                                                                                                   | 0-1         | 1       | -    | U16       | At stop           | Immediately      |
| 0Fh   | COA.0E    | Commissioning software data format                 | 0: Low 16 bits before high 16 bits<br>1: High 16 bits before low 16 bits                                                                                                      | 0-1         | 0       | -    | U16       | At stop           | Immediately      |
| 13h   | COA.12    | USB display switch                                 | -                                                                                                                                                                             | 0-1         | 1       | -    | U16       | At stop           | Upon re-power-on |

## NOTICE

- For details about parameters above, refer to section 8.24.5 "Group COA".

## 8.10 Setting of Position Comparison (200Dh/C0D)

| Index | Parameter | Name                                     | Options                                                                                                                                                                                                                                                                                                          | Value Range | Default | Unit  | Data Type | Modification Mode | Effective Time   |
|-------|-----------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------|-----------|-------------------|------------------|
| 01h   | C0D.00    | Position comparison output switch        | 0: Disabled<br>1: Enabled                                                                                                                                                                                                                                                                                        | 0-1         | 0       | -     | U16       | During operation  | Immediately      |
| 03h   | C0D.02    | Position comparison resolution           | 0: 24bit<br>1: 23bit<br>2: 22bit<br>3: 21bit<br>4: 20bit<br>5: 19bit<br>6: 18bit<br>7: 17bit                                                                                                                                                                                                                     | 0-7         | 1       | -     | U16       | During operation  | Immediately      |
| 04h   | C0D.03    | Position comparison mode                 | 0: Individual absolute comparison mode<br>1: Cyclic incremental comparison mode<br>2: Fixed cyclic incremental mode<br>3: Cyclic absolute comparison mode                                                                                                                                                        | 0-3         | 0       | -     | U16       | During operation  | Immediately      |
| 06h   | C0D.05    | Position comparison output width         | -                                                                                                                                                                                                                                                                                                                | 5-1200      | 10      | 0.1ms | U16       | During operation  | Upon re-power-on |
| 07h   | C0D.06    | Position comparison output port polarity | Bit00: Channel 1 polarity<br>0: Negative polarity<br>1: Positive polarity<br>Bit01: Channel 2 polarity<br>0: Negative polarity<br>1: Positive polarity<br>Bit02: Channel 3 polarity<br>0: Negative polarity<br>1: Positive polarity<br>Bit03: Channel 4 polarity<br>0: Negative polarity<br>1: Positive polarity | 0-65535     | 15      | -     | U16       | During operation  | Upon re-power-on |
| 08h   | C0D.07    | Start point of position comparison       | -                                                                                                                                                                                                                                                                                                                | 1-40        | 1       | -     | U16       | During operation  | Immediately      |
| 09h   | C0D.08    | End point of position comparison         | -                                                                                                                                                                                                                                                                                                                | 1-40        | 1       | -     | U16       | During operation  | Immediately      |
| 0Ah   | C0D.09    | Position comparison status               | -                                                                                                                                                                                                                                                                                                                | 0-1024      | 0       | -     | U16       | Read only         | Immediately      |

| Index | Parameter | Name                                               | Options | Value Range            | Default | Unit   | Data Type | Modification Mode | Effective Time   |
|-------|-----------|----------------------------------------------------|---------|------------------------|---------|--------|-----------|-------------------|------------------|
| 08h   | COD.0A    | Real-time position of position comparison          | -       | $-2^{31} - (2^{31}-1)$ | 0       | -      | I32       | Read only         | Immediately      |
| 0Dh   | COD.0C    | Zero offset of position comparison                 | -       | $-2^{31} - (2^{31}-1)$ | 0       | -      | I32       | During operation  | Immediately      |
| 0Fh   | COD.0E    | Compensation for position comparison output de-lay | -       | 0-10000                | 0       | 0.01us | U16       | At stop           | Upon re-power-on |
| 10h   | COD.0F    | Fixed number of cycles                             | -       | 1-65535                | 1       | -      | U16       | During operation  | Immediately      |
| 12h   | COD.11    | Number of cycles in certain mode                   | -       | 0-65535                | 0       | -      | U16       | Read only         | Immediately      |
| 14h   | COD.13    | Start comparison point in loop absolute mode       | -       | 1-40                   | 1       | -      | U16       | During operation  | Immediately      |

## 8.11 Position Comparison Parameters (200Eh/C0E)

| Index | Parameter | Name                           | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 01h   | C0E.00    | Setting of target position 1   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 03h   | C0E.02    | Attribute of target position 1 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 04h   | C0E.03    | Setting of target position 2   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                           | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 06h   | COE.05    | Attribute of target position 2 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 07h   | COE.06    | Setting of target position 3   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 09h   | COE.08    | Attribute of target position 3 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 0Ah   | COE.09    | Setting of target position 4   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 0Ch   | COE.0B    | Attribute of target position 4 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |



| Index | Parameter | Name                           | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 0Dh   | COE.0C    | Setting of target position 5   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 0Fh   | COE.0E    | Attribute of target position 5 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 10h   | COE.0F    | Setting of target position 6   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 12h   | COE.11    | Attribute of target position 6 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 13h   | COE.12    | Setting of target position 7   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                           | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 15h   | COE.14    | Attribute of target position 7 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 16h   | COE.15    | Setting of target position 8   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 18h   | COE.17    | Attribute of target position 8 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 19h   | COE.18    | Setting of target position 9   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 18h   | COE.1A    | Attribute of target position 9 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 1Ch   | COE.1B    | Setting of target position 10   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 1Eh   | COE.1D    | Attribute of target position 10 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 1Fh   | COE.1E    | Setting of target position 11   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 21h   | COE.20    | Attribute of target position 11 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 22h   | COE.21    | Setting of target position 12   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range                  | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|-----------|-------------------|----------------|
| 24h   | COE.23    | Attribute of target position 12 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 25h   | COE.24    | Setting of target position 13   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 27h   | COE.26    | Attribute of target position 13 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 28h   | COE.27    | Setting of target position 14   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 2Ah   | COE.29    | Attribute of target position 14 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 2Bh   | C0E.2A    | Setting of target position 15   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 2Dh   | C0E.2C    | Attribute of target position 15 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 2Eh   | C0E.2D    | Setting of target position 16   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 30h   | C0E.2F    | Attribute of target position 16 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 31h   | C0E.30    | Setting of target position 17   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 33h   | C0E.32    | Attribute of target position 17 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 34h   | C0E.33    | Setting of target position 18   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 36h   | C0E.35    | Attribute of target position 18 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 37h   | C0E.36    | Setting of target position 19   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 39h   | C0E.38    | Attribute of target position 19 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 3Ah   | C0E.39    | Setting of target position 20   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 3Ch   | C0E.3B    | Attribute of target position 20 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 3Dh   | C0E.3C    | Setting of target position 21   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 3Fh   | C0E.3E    | Attribute of target position 21 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 40h   | C0E.3F    | Setting of target position 22   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range                  | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|-----------|-------------------|----------------|
| 42h   | C0E.41    | Attribute of target position 22 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 43h   | C0E.42    | Setting of target position 23   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 45h   | C0E.44    | Attribute of target position 23 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 46h   | C0E.45    | Setting of target position 24   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 48h   | C0E.47    | Attribute of target position 24 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |



| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 49h   | C0E.48    | Setting of target position 25   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 4Bh   | C0E.4A    | Attribute of target position 25 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 4Ch   | C0E.4B    | Setting of target position 26   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 4Eh   | C0E.4D    | Attribute of target position 26 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 4Fh   | C0E.4E    | Setting of target position 27   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 51h   | C0E.50    | Attribute of target position 27 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 52h   | C0E.51    | Setting of target position 28   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 54h   | C0E.53    | Attribute of target position 28 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 55h   | C0E.54    | Setting of target position 29   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 57h   | C0E.56    | Attribute of target position 29 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 58h   | C0E.57    | Setting of target position 30   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 5Ah   | C0E.59    | Attribute of target position 30 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 5Bh   | C0E.5A    | Setting of target position 31   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 5Dh   | C0E.5C    | Attribute of target position 31 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 5Eh   | C0E.5D    | Setting of target position 32   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range            | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|------|-----------|-------------------|----------------|
| 60h   | C0E.5F    | Attribute of target position 32 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 61h   | C0E.60    | Setting of target position 33   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 63h   | C0E.62    | Attribute of target position 33 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |
| 64h   | C0E.63    | Setting of target position 34   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31}-1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 66h   | C0E.65    | Attribute of target position 34 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range              | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|------|-----------|-------------------|----------------|
| 67h   | C0E.66    | Setting of target position 35   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 69h   | C0E.68    | Attribute of target position 35 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 6Ah   | C0E.69    | Setting of target position 36   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |
| 6Ch   | C0E.6B    | Attribute of target position 36 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                  | 0       | -    | U16       | During operation  | Immediately    |
| 6Dh   | C0E.6C    | Setting of target position 37   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31} - (2^{31} - 1)$ | 0       | -    | I32       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range                  | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|-----------|-------------------|----------------|
| 6Fh   | C0E.6E    | Attribute of target position 37 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 70h   | C0E.6F    | Setting of target position 38   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 72h   | C0E.71    | Attribute of target position 38 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |
| 73h   | C0E.72    | Setting of target position 39   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 75h   | C0E.74    | Attribute of target position 39 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |

| Index | Parameter | Name                            | Options                                                                                                                                                                                                                                                                                                                                        | Value Range                  | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|-----------|-------------------|----------------|
| 76h   | C0E.75    | Setting of target position 40   | -                                                                                                                                                                                                                                                                                                                                              | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | -    | I32       | During operation  | Immediately    |
| 78h   | C0E.77    | Attribute of target position 40 | Bit00: Output DO active signal if current position changes from "less than" to "more than" the comparison point<br>Bit01: Output DO active signal if current position changes from "more than" to "less than" the comparison point<br>Bit08: Channel 1 output<br>Bit09: Channel 2 output<br>Bit10: Channel 3 output<br>Bit11: Channel 4 output | 0-65535                      | 0       | -    | U16       | During operation  | Immediately    |

## 8.12 Homing Touch Probe Parameters (2010h/C10)

| Index | Parameter | Name                     | Options                                                                                         | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------|-------------------------------------------------------------------------------------------------|----------------|---------|------|-----------|-------------------|----------------|
| 01h   | C10.00    | Homing enable            | 0: Inactive<br>1: Written through communication<br>2: DI trigger<br>3: Current position as home | 0-3            | 0       | -    | U16       | During operation  | Immediately    |
| 02h   | C10.01    | Homing method selection  | -                                                                                               | -2-35          | 0       | -    | I16       | During operation  | Immediately    |
| 03h   | C10.02    | Initial homing speed     | -                                                                                               | 0-8000         | 100     | rpm  | U16       | During operation  | Immediately    |
| 04h   | C10.03    | End homing speed         | -                                                                                               | 0-3000         | 10      | rpm  | U16       | During operation  | Immediately    |
| 05h   | C10.04    | Homing acceleration time | -                                                                                               | 0-3600000      | 1000    | ms   | U32       | During operation  | Immediately    |
| 07h   | C10.06    | Homing deceleration time | -                                                                                               | 0-3600000      | 1000    | ms   | U32       | During operation  | Immediately    |
| 09h   | C10.08    | Homing timeout interval  | -                                                                                               | $0-(2^{32}-1)$ | 60000   | ms   | U32       | During operation  | Immediately    |
| 08h   | C10.0A    | Homing offset mode       | 0: Absolute position<br>1: Relative position                                                    | 0-1            | 0       | -    | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                                                         | Options                                                                                                                              | Value Range    | Default | Unit                | Data Type | Modification Mode | Effective Time   |
|-------|-----------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------|---------|---------------------|-----------|-------------------|------------------|
| 0Ch   | C10.0B    | Homing offset distance                                                       | -                                                                                                                                    | $-2^{31}-1$    | 0       | Unit in application | I32       | During operation  | Immediately      |
| 11h   | C10.10    | Multi-turn absolute position offset (low 32 bits)                            | -                                                                                                                                    | $-2^{31}-1$    | 0       | P                   | I32       | At stop           | Upon re-power-on |
| 13h   | C10.12    | Multi-turn absolute position offset (high 32 bits)                           | -                                                                                                                                    | $-2^{31}-1$    | 0       | P                   | I32       | At stop           | Upon re-power-on |
| 15h   | C10.14    | Multi-turn revolutions data offset                                           | -                                                                                                                                    | 0-65535        | 0       | Rev                 | U16       | Read only         | Immediately      |
| 16h   | C10.15    | Multi-turn overflow flag                                                     | -                                                                                                                                    | 0-1            | 0       | -                   | U16       | Read only         | Immediately      |
| 17h   | C10.16    | Reference running mode in rotation mode                                      | 0: Nearest<br>1: Always in forward direction<br>2: Always in reverse direction<br>3: Always in current direction<br>4: Not specified | 0-4            | 0       | -                   | U16       | At stop           | Immediately      |
| 19h   | C10.18    | Numerator of electronic gear ratio in rotation mode                          | -                                                                                                                                    | 1-65535        | 1       | -                   | U16       | At stop           | Immediately      |
| 1Ah   | C10.19    | Denominator of electronic gear ratio in rotation mode                        | -                                                                                                                                    | 1-65535        | 1       | -                   | U16       | At stop           | Immediately      |
| 18h   | C10.1A    | Upper limit of mechanical absolute position in ro-tation mode (low 32 bits)  | -                                                                                                                                    | $0-(2^{32}-1)$ | 0       | P                   | U32       | At stop           | Immediately      |
| 1Dh   | C10.1C    | Upper limit of mechanical absolute position in ro-tation mode (high 32 bits) | -                                                                                                                                    | $0-(2^{32}-1)$ | 0       | P                   | U32       | At stop           | Immediately      |
| 1Fh   | C10.1E    | Single-turn homing absolute value offset                                     | -                                                                                                                                    | $-2^{31}-1$    | 0       | Unit in application | I32       | At stop           | Upon re-power-on |
| 21h   | C10.20    | Probe capture enable                                                         | 0: Inactive<br>1: Enabled                                                                                                            | 0-1            | 0       | -                   | U16       | At stop           | Upon re-power-on |



| Index | Parameter | Name                                                   | Options                                          | Value Range                  | Default | Unit                | Data Type | Modification Mode | Effective Time   |
|-------|-----------|--------------------------------------------------------|--------------------------------------------------|------------------------------|---------|---------------------|-----------|-------------------|------------------|
| 22h   | C10.21    | Release of probe capture state                         | 0: Do not release<br>1: Release                  | 0-1                          | 0       | -                   | U16       | At stop           | Upon re-power-on |
| 23h   | C10.22    | Probe capture offset distance                          | -                                                | $-2^{31}-$<br>( $2^{31}-1$ ) | 0       | Unit in application | I32       | During operation  | Immediately      |
| 25h   | C10.24    | Probe capture speed                                    | -                                                | 0-8000                       | 100     | rpm                 | U16       | During operation  | Immediately      |
| 26h   | C10.25    | Probe capture acceleration time                        | -                                                | 0-3600000                    | 10      | ms                  | U32       | During operation  | Immediately      |
| 28h   | C10.27    | Probe capture deceleration time                        | -                                                | 0-3600000                    | 10      | ms                  | U16       | During operation  | Immediately      |
| 2Dh   | C10.2C    | Probe capture pin filter time                          | -                                                | 0-10200                      | 200     | 0.01us              | U16       | At stop           | Upon re-power-on |
| 31h   | C10.30    | Torque limit of homing upon hit-and-stop               | -                                                | 0-3000                       | 1000    | 0.1%                | U16       | During operation  | Immediately      |
| 32h   | C10.31    | Speed for homing upon hit-and-stop                     | -                                                | 5-1000                       | 10      | rpm                 | U16       | During operation  | Immediately      |
| 33h   | C10.32    | Number of times for homing upon hit-and-stop           | -                                                | 5-65535                      | 30      | -                   | U16       | During operation  | Immediately      |
| 34h   | C10.33    | Positioning threshold equivalent upon homing completed | -                                                | 0-50                         | 5       | rpm                 | U16       | At stop           | Upon re-power-on |
| 35h   | C10.34    | Mode of homing upon hit-and-stop                       | 0: Deviation not cleared<br>1: Deviation cleared | 0-1                          | 0       | -                   | U16       | During operation  | Immediately      |

### 8.13 Position planning (2011h/C11)

| Index | Parameter | Name                                               | Options                                                                                                                                         | Value Range            | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|---------------------|-----------|-------------------|----------------|
| 01h   | C11.00    | Position planning mode selection                   | 0: Single operation<br>1: Cyclic operation<br>2: DI selection mode<br>3: PP mode<br>4: Single continuous operation<br>5: Cyclic continuous mode | 0-5                    | 0       | -                   | U16       | At stop           | Immediately    |
| 02h   | C11.01    | Position planning reference type                   | 0: Absolute reference<br>1: Relative reference                                                                                                  | 0-1                    | 0       | -                   | U16       | At stop           | Immediately    |
| 03h   | C11.02    | Position planning reference update mode            | 0: Cache update<br>1: Update now                                                                                                                | 0-1                    | 0       | -                   | U16       | At stop           | Immediately    |
| 04h   | C11.03    | Position planning initial group number             | -                                                                                                                                               | 1-16                   | 1       | -                   | U16       | At stop           | Immediately    |
| 05h   | C11.04    | Position planning end group number                 | -                                                                                                                                               | 1-16                   | 1       | -                   | U16       | At stop           | Immediately    |
| 06h   | C11.05    | Processing of position planning remaining segments | 0: Continue running the unfinished segment<br>1: Restart running from the first segment                                                         | 0-1                    | 0       | -                   | U16       | At stop           | Immediately    |
| 07h   | C11.06    | Group 1 planning displacement                      | -                                                                                                                                               | $-2^{31} - (2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 09h   | C11.08    | Group 1 planning speed                             | -                                                                                                                                               | 0-8000                 | 100     | rpm                 | U16       | During operation  | Immediately    |
| 08h   | C11.0A    | Group 1 planning acceleration time                 | -                                                                                                                                               | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| 0Dh   | C11.0C    | Group 1 planning deceleration time                 | -                                                                                                                                               | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| 0Fh   | C11.0E    | Group 1 planning waiting time                      | -                                                                                                                                               | $0-(2^{32}-1)$         | 10      | ms                  | U32       | During operation  | Immediately    |
| 11h   | C11.10    | Group 2 planning displacement                      | -                                                                                                                                               | $-2^{31} - (2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 13h   | C11.12    | Group 2 planning speed                             | -                                                                                                                                               | 0-8000                 | 100     | rpm                 | U16       | During operation  | Immediately    |
| 15h   | C11.14    | Group 2 planning acceleration time                 | -                                                                                                                                               | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |

| Index | Parameter | Name                               | Options | Value Range          | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------------------|---------|----------------------|---------|---------------------|-----------|-------------------|----------------|
| 17h   | C11.16    | Group 2 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 19h   | C11.18    | Group 2 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 21h   | C11.20    | Group 3 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 23h   | C11.22    | Group 3 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 25h   | C11.24    | Group 3 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 27h   | C11.26    | Group 3 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 29h   | C11.28    | Group 3 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 31h   | C11.30    | Group 4 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 33h   | C11.32    | Group 4 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 35h   | C11.34    | Group 4 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 37h   | C11.36    | Group 4 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 39h   | C11.38    | Group 4 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 41h   | C11.40    | Group 5 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 43h   | C11.42    | Group 5 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 45h   | C11.44    | Group 5 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 47h   | C11.46    | Group 5 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 49h   | C11.48    | Group 5 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 51h   | C11.50    | Group 6 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 53h   | C11.52    | Group 6 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |

| Index | Parameter | Name                               | Options | Value Range          | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|------------------------------------|---------|----------------------|---------|---------------------|-----------|-------------------|----------------|
| 55h   | C11.54    | Group 6 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 57h   | C11.56    | Group 6 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 59h   | C11.58    | Group 6 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 61h   | C11.60    | Group 7 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 63h   | C11.62    | Group 7 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 65h   | C11.64    | Group 7 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 67h   | C11.66    | Group 7 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 69h   | C11.68    | Group 7 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 71h   | C11.70    | Group 8 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 73h   | C11.72    | Group 8 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 75h   | C11.74    | Group 8 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 77h   | C11.76    | Group 8 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 79h   | C11.78    | Group 8 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 81h   | C11.80    | Group 9 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| 83h   | C11.82    | Group 9 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 85h   | C11.84    | Group 9 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 87h   | C11.86    | Group 9 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 89h   | C11.88    | Group 9 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| 91h   | C11.90    | Group 10 planning displacement     | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |

| Index | Parameter | Name                                | Options | Value Range          | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------|---------|----------------------|---------|---------------------|-----------|-------------------|----------------|
| 93h   | C11.92    | Group 10 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| 95h   | C11.94    | Group 10 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 97h   | C11.96    | Group 10 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| 99h   | C11.98    | Group 10 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| A1h   | C11.A0    | Group 11 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| A3h   | C11.A2    | Group 11 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| A5h   | C11.A4    | Group 11 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| A7h   | C11.A6    | Group 11 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| A9h   | C11.A8    | Group 11 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| B1h   | C11.B0    | Group 12 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| B3h   | C11.B2    | Group 12 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| B5h   | C11.B4    | Group 12 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| B7h   | C11.B6    | Group 12 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| B9h   | C11.B8    | Group 12 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |
| C1h   | C11.C0    | Group 13 planning displacement      | -       | $-2^{31}-(2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| C3h   | C11.C2    | Group 13 planning speed             | -       | 0-8000               | 100     | rpm                 | U16       | During operation  | Immediately    |
| C5h   | C11.C4    | Group 13 planning acceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| C7h   | C11.C6    | Group 13 planning deceleration time | -       | 0-65535              | 10      | ms                  | U32       | During operation  | Immediately    |
| C9h   | C11.C8    | Group 13 planning waiting time      | -       | $0-(2^{32}-1)$       | 10      | ms                  | U32       | During operation  | Immediately    |

| Index | Parameter | Name                                | Options | Value Range            | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------|---------|------------------------|---------|---------------------|-----------|-------------------|----------------|
| D1h   | C11.D0    | Group 14 planning displacement      | -       | $-2^{31} - (2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| D3h   | C11.D2    | Group 14 planning speed             | -       | 0-8000                 | 100     | rpm                 | U32       | During operation  | Immediately    |
| D5h   | C11.D4    | Group 14 planning acceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| D7h   | C11.D6    | Group 14 planning deceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| D9h   | C11.D8    | Group 14 planning waiting time      | -       | $0-(2^{32}-1)$         | 10      | ms                  | U32       | During operation  | Immediately    |
| E1h   | C11.E0    | Group 15 planning displacement      | -       | $-2^{31} - (2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| E3h   | C11.E2    | Group 15 planning speed             | -       | 0-8000                 | 100     | rpm                 | U16       | During operation  | Immediately    |
| E5h   | C11.E4    | Group 15 planning acceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| E7h   | C11.E6    | Group 15 planning deceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| E9h   | C11.E8    | Group 15 planning waiting time      | -       | $0-(2^{32}-1)$         | 10      | ms                  | U32       | During operation  | Immediately    |
| F1h   | C11.F0    | Group 16 planning displacement      | -       | $-2^{31} - (2^{31}-1)$ | 10000   | Unit in application | I32       | During operation  | Immediately    |
| F3h   | C11.F2    | Group 16 planning speed             | -       | 0-8000                 | 100     | rpm                 | U16       | During operation  | Immediately    |
| F5h   | C11.F4    | Group 16 planning acceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| F7h   | C11.F6    | Group 16 planning deceleration time | -       | 0-65535                | 10      | ms                  | U32       | During operation  | Immediately    |
| F9h   | C11.F8    | Group 16 planning waiting time      | -       | $0-(2^{32}-1)$         | 10      | ms                  | U32       | During operation  | Immediately    |

## 8.14 Speed planning (2012h/C12)

| Index | Parameter | Name                          | Options                                                                 | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------|-------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | C12.00    | Speed planning mode selection | 0: Single operation<br>1: Cyclic operation<br>2: DI selection operation | 0-2         | 0       | -    | U16       | At stop           | Immediately    |

| Index | Parameter | Name                                | Options | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------|---------|----------------|---------|------|-----------|-------------------|----------------|
| 02h   | C12.01    | Speed planning initial group number | -       | 1-16           | 1       | -    | U16       | At stop           | Immediately    |
| 03h   | C12.02    | Speed planning end group number     | -       | 1-16           | 1       | -    | U16       | At stop           | Immediately    |
| 07h   | C12.06    | Group 1 speed reference             | -       | -8000-8000     | 100     | rpm  | I16       | During operation  | Immediately    |
| 09h   | C12.08    | Group 1 running time                | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 08h   | C12.0A    | Group 1 acceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 0Dh   | C12.0C    | Group 1 deceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 11h   | C12.10    | Group 2 speed reference             | -       | -8000-8000     | 500     | rpm  | I16       | During operation  | Immediately    |
| 13h   | C12.12    | Group 2 running time                | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 15h   | C12.14    | Group 2 acceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 17h   | C12.16    | Group 2 deceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 21h   | C12.20    | Group 3 speed reference             | -       | -8000-8000     | 1000    | rpm  | I16       | During operation  | Immediately    |
| 23h   | C12.22    | Group 3 running time                | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 25h   | C12.24    | Group 3 acceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 27h   | C12.26    | Group 3 deceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 31h   | C12.30    | Group 4 speed reference             | -       | -8000-8000     | 1500    | rpm  | I16       | During operation  | Immediately    |
| 33h   | C12.32    | Group 4 running time                | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 35h   | C12.34    | Group 4 acceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 37h   | C12.36    | Group 4 deceleration time           | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 41h   | C12.40    | Group 5 speed reference             | -       | -8000-8000     | -1500   | rpm  | I16       | During operation  | Immediately    |
| 43h   | C12.42    | Group 5 running time                | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |

| Index | Parameter | Name                      | Options | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------|---------|----------------|---------|------|-----------|-------------------|----------------|
| 45h   | C12.44    | Group 5 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 47h   | C12.46    | Group 5 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 51h   | C12.50    | Group 6 speed reference   | -       | -8000-8000     | -1000   | rpm  | I16       | During operation  | Immediately    |
| 53h   | C12.52    | Group 6 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 55h   | C12.54    | Group 6 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 57h   | C12.56    | Group 6 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 61h   | C12.60    | Group 7 speed reference   | -       | -8000-8000     | -500    | rpm  | I16       | During operation  | Immediately    |
| 63h   | C12.62    | Group 7 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 65h   | C12.64    | Group 7 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 67h   | C12.66    | Group 7 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 71h   | C12.70    | Group 8 speed reference   | -       | -8000-8000     | -100    | rpm  | I16       | During operation  | Immediately    |
| 73h   | C12.72    | Group 8 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 75h   | C12.74    | Group 8 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 77h   | C12.76    | Group 8 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 81h   | C12.80    | Group 9 speed reference   | -       | -8000-8000     | 100     | rpm  | I16       | During operation  | Immediately    |
| 83h   | C12.82    | Group 9 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| 85h   | C12.84    | Group 9 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 87h   | C12.86    | Group 9 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 91h   | C12.90    | Group 10 speed reference  | -       | -8000-8000     | 500     | rpm  | I16       | During operation  | Immediately    |
| 93h   | C12.92    | Group 10 running time     | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |



| Index | Parameter | Name                       | Options | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------|---------|----------------|---------|------|-----------|-------------------|----------------|
| 95h   | C12.94    | Group 10 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| 97h   | C12.96    | Group 10 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| A1h   | C12.A0    | Group 11 speed reference   | -       | -8000-8000     | 1000    | rpm  | I16       | During operation  | Immediately    |
| A3h   | C12.A2    | Group 11 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| A5h   | C12.A4    | Group 11 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| A7h   | C12.A6    | Group 11 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| B1h   | C12.B0    | Group 12 speed reference   | -       | -8000-8000     | 1500    | rpm  | I16       | During operation  | Immediately    |
| B3h   | C12.B2    | Group 12 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| B5h   | C12.B4    | Group 12 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| B7h   | C12.B6    | Group 12 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| C1h   | C12.C0    | Group 13 speed reference   | -       | -8000-8000     | -1500   | rpm  | I16       | During operation  | Immediately    |
| C3h   | C12.C2    | Group 13 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| C5h   | C12.C4    | Group 13 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| C7h   | C12.C6    | Group 13 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| D1h   | C12.D0    | Group 14 speed reference   | -       | -8000-8000     | -1000   | rpm  | I16       | During operation  | Immediately    |
| D3h   | C12.D2    | Group 14 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| D5h   | C12.D4    | Group 14 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| D7h   | C12.D6    | Group 14 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| E1h   | C12.E0    | Group 15 speed reference   | -       | -8000-8000     | -500    | rpm  | I16       | During operation  | Immediately    |
| E3h   | C12.E2    | Group 15 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |

| Index | Parameter | Name                       | Options | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------|---------|----------------|---------|------|-----------|-------------------|----------------|
| E5h   | C12.E4    | Group 15 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| E7h   | C12.E6    | Group 15 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| F1h   | C12.F0    | Group 16 speed reference   | -       | -8000-8000     | -100    | rpm  | I16       | During operation  | Immediately    |
| F3h   | C12.F2    | Group 16 running time      | -       | $0-(2^{32}-1)$ | 10000   | ms   | U32       | During operation  | Immediately    |
| F5h   | C12.F4    | Group 16 acceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |
| F7h   | C12.F6    | Group 16 deceleration time | -       | 0-65535        | 10      | ms   | U32       | During operation  | Immediately    |

## 8.15 Fully closed-loop (201Bh/C1B)

| Index | Parameter | Name                                                          | Options                                                                                  | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|---------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------|---------|------|-----------|-------------------|------------------|
| 01h   | C1B.00    | Encoder feedback mode                                         | 0: Internal encoder<br>1: External encoder<br>2: Switching between inner and outer loops | 0-2            | 0       | -    | U16       | At stop           | Immediately      |
| 02h   | C1B.01    | Use mode of external encoder                                  | 0: Normal count direction<br>1: Count direction reversed                                 | 0-1            | 0       | -    | U16       | At stop           | Immediately      |
| 03h   | C1B.02    | Selection of absolute value mode of external encoder          | 0: Incremental mode<br>1: Absolute mode                                                  | 0-1            | 0       | -    | U16       | At stop           | Upon re-power-on |
| 04h   | C1B.03    | Selection of feedback type of external encoder                | 0: ABZ encoder<br>1: BISS-C encoder                                                      | 0-1            | 0       | -    | U16       | At stop           | Upon re-power-on |
| 05h   | C1B.04    | Number of external encoder pulses for one revolution of motor | -                                                                                        | $0-(2^{31}-1)$ | 10000   | P    | I32       | At stop           | Upon re-power-on |
| 09h   | C1B.08    | Maximum deviation of mixed control                            | -                                                                                        | $0-(2^{31}-1)$ | 1000    | P    | I32       | At stop           | Immediately      |

| Index | Parameter | Name                                                                 | Options                   | Value Range                   | Default | Unit  | Data Type | Modification Mode | Effective Time   |
|-------|-----------|----------------------------------------------------------------------|---------------------------|-------------------------------|---------|-------|-----------|-------------------|------------------|
| 0Bh   | C1B.0A    | Mixed control deviation clearance lap number setting                 | -                         | 0-100                         | 1       | -     | U16       | During operation  | Immediately      |
| 0Ch   | C1B.0B    | First-order low-pass filter time of external and internal deviations | -                         | 0-65535                       | 0       | 0.1ms | U16       | At stop           | Immediately      |
| 0Fh   | C1B.0E    | Fully closed-loop external position error                            | -                         | $-2^{31}$ -<br>( $2^{31}-1$ ) | 0       | P     | I32       | Read only         | Immediately      |
| 11h   | C1B.10    | Internal encoder feedback value                                      | -                         | $-2^{31}$ -<br>( $2^{31}-1$ ) | 0       | P     | I32       | Read only         | Immediately      |
| 13h   | C1B.12    | External encoder feedback value                                      | -                         | $-2^{31}$ -<br>( $2^{31}-1$ ) | 0       | P     | I32       | Read only         | Immediately      |
| 18h   | C1B.1A    | BISS encoder feedback position                                       | -                         | $-2^{31}$ -<br>( $2^{31}-1$ ) | 0       | -     | I32       | Read only         | Immediately      |
| 1Dh   | C1B.1C    | BISS communication alarm index value                                 | -                         | 0-65535                       | 0       | -     | U16       | Read only         | Immediately      |
| 1Eh   | C1B.1D    | BISS communication position feedback active bit                      | -                         | 16-48                         | 29      | -     | U16       | At stop           | Upon re-power-on |
| 1Fh   | C1B.1E    | Selection of baud rate setting for BISS communication                | 0: 1.6M<br>1: 2M<br>2: 4M | 0-2                           | 1       | -     | U16       | At stop           | Upon re-power-on |

## 8.16 Motor Parameters (2020h/R20)

| Index | Parameter | Name        | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|-------------|---------|-------------|---------|------|-----------|-------------------|------------------|
| 01h   | R20.00    | Motor model | -       | 0-65535     | 10000   | -    | U16       | At stop           | Upon re-power-on |
| 04h   | R20.03    | Motor SN 1  | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately      |
| 05h   | R20.04    | Motor SN 2  | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately      |
| 06h   | R20.05    | Motor SN 3  | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately      |

| Index | Parameter | Name                     | Options                      | Value Range    | Default | Unit   | Data Type | Modification Mode | Effective Time   |
|-------|-----------|--------------------------|------------------------------|----------------|---------|--------|-----------|-------------------|------------------|
| 07h   | R20.06    | Motor SN 4               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |
| 08h   | R20.07    | Motor SN 5               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |
| 09h   | R20.08    | Rated motor voltage      | 0: 220V<br>1: 380V<br>2: 48V | 0-2            | 0       | -      | U16       | At stop           | Upon re-power-on |
| 0Ah   | R20.09    | Rated motor power        | -                            | $1-(2^{32}-1)$ | 40      | 0.01kW | U32       | At stop           | Upon re-power-on |
| 0Ch   | R20.0B    | Rated motor current      | -                            | $1-(2^{32}-1)$ | 250     | 0.01A  | U32       | At stop           | Upon re-power-on |
| 21h   | R20.20    | Encoder software version | -                            | 0-65535        | 0       | 0.1    | U16       | Read only         | Immediately      |
| 23h   | R20.22    | Encoder type             | -                            | 0-65535        | 0       | -      | U16       | Read only         | Upon re-power-on |
| 26h   | R20.25    | Encoder initial position | -                            | $0-(2^{32}-1)$ | 0       | Unit   | U32       | At stop           | Upon re-power-on |
| 28h   | R20.2A    | Encoder resolution bit   | -                            | 0-32           | 23      | -      | U16       | At stop           | Upon re-power-on |
| 41h   | R20.40    | Motor SN 6               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |
| 42h   | R20.41    | Motor SN 7               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |
| 43h   | R20.42    | Motor SN 8               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |
| 44h   | R20.43    | Motor SN 9               | -                            | 0-65535        | 0       | -      | U16       | Read only         | Immediately      |

## 8.17 Drive Parameters (2021h/R21)

| Index | Parameter | Name                | Options | Value Range    | Default | Unit   | Data Type | Modification Mode | Effective Time   |
|-------|-----------|---------------------|---------|----------------|---------|--------|-----------|-------------------|------------------|
| 01h   | R21.00    | Drive model         | -       | 0-65535        | 3       | -      | U16       | At stop           | Upon re-power-on |
| 0Dh   | R21.0C    | Drive voltage class | -       | 0-2            | 0       | -      | U16       | Read only         | Immediately      |
| 0Eh   | R21.0D    | Rated drive power   | -       | $1-(2^{32}-1)$ | 40      | 0.01kW | U32       | Read only         | Immediately      |

| Index | Parameter | Name                                                       | Options | Value Range    | Default | Unit               | Data Type | Modification Mode | Effective Time   |
|-------|-----------|------------------------------------------------------------|---------|----------------|---------|--------------------|-----------|-------------------|------------------|
| 10h   | R21.0F    | Rated output current of drive                              | -       | $1-(2^{32}-1)$ | 280     | 0.01A              | U32       | Read only         | Immediately      |
| 12h   | R21.11    | Maximum output current of drive                            | -       | $1-(2^{32}-1)$ | 1010    | 0.01A              | U32       | Read only         | Immediately      |
| 14h   | R21.13    | Internal bleeder resistor power                            | -       | 1-65535        | 1       | W                  | U16       | Read only         | Immediately      |
| 15h   | R21.14    | Internal bleeder resistor resistance                       | -       | 1-65535        | 1       | $\Omega$           | U16       | Read only         | Immediately      |
| 16h   | R21.15    | Minimum resistance of external bleeder resistor            | -       | 0-65535        | 40      | $\Omega$           | U16       | Read only         | Immediately      |
| 18h   | R21.17    | Scaling coefficient for bus voltage of the control circuit | -       | 0-1500         | 1000    | 0.1%               | U16       | At stop           | Immediately      |
| 19h   | R21.18    | Bus voltage scaling coefficient                            | -       | 500-1500       | 1000    | 0.1%               | U16       | At stop           | Immediately      |
| 1Ah   | R21.19    | Drive undervoltage threshold                               | -       | 0-10000        | 2000    | 0.1V               | U16       | At stop           | Upon re-power-on |
| 18h   | R21.1A    | Drive overvoltage threshold                                | -       | 0-10000        | 4200    | 0.1V               | U16       | At stop           | Upon re-power-on |
| 1Ch   | R21.1B    | Drive bleeder threshold                                    | -       | 0-10000        | 3800    | 0.1V               | U16       | At stop           | Upon re-power-on |
| 41h   | R21.40    | NTC temperature warning threshold                          | -       | 0-120          | 96      | $^{\circ}\text{C}$ | U16       | At stop           | Immediately      |

### NOTICE

- For details about parameters above, refer to section 8.24.6 "Group R21".

## 8.18 Motor Gain Parameters (2022h/R22)

| Index | Parameter | Name                        | Options                                 | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time   |
|-------|-----------|-----------------------------|-----------------------------------------|-------------|---------|------|-----------|-------------------|------------------|
| 01h   | R22.00    | Current loop mode           | 0: Standard mode<br>1: Performance mode | 0-1         | 0       | -    | U16       | At stop           | Upon re-power-on |
| 02h   | R22.01    | Current loop response level | -                                       | 0-4000      | 0       | 0.1% | U16       | At stop           | Upon re-power-on |

| Index | Parameter | Name                                               | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 21h   | R22.20    | MTPA field-weakening switch                        | -       | 0-65535     | 256     | -    | U16       | At stop           | Immediately    |
| 22h   | R22.21    | Field-weakening depth                              | -       | 500-2000    | 1000    | 0.1% | U16       | At stop           | Immediately    |
| 23h   | R22.22    | Field-weakening proportional gain                  | -       | 10-1000     | 100     | Hz   | U16       | At stop           | Immediately    |
| 24h   | R22.23    | Field-weakening integral gain                      | -       | 0-8000      | 50      | 0.1% | U16       | At stop           | Immediately    |
| 25h   | R22.24    | Cutoff frequency of d axis current low-pass filter | -       | 0-16000     | 0       | Hz   | U16       | At stop           | Immediately    |
| 26h   | R22.25    | Field-weakening d axis current limit               | -       | 0-3000      | 1500    | 0.1% | U16       | At stop           | Immediately    |
| 29h   | R22.28    | Overmodulation ratio                               | -       | 900-1155    | 1155    | 0.1% | U16       | At stop           | Immediately    |
| 31h   | R22.30    | Dead zone compensation                             | -       | 0-2000      | 1000    | 0.1% | U16       | At stop           | Immediately    |

## 8.19 Parameters of Control in Progress (2030h/F30)

| Index | Parameter | Name                                 | Options                               | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|--------------------------------------|---------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 11h   | F30.10    | Inertia auto-tuning selection        | 0: Disabled<br>1: Enabled             | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |
| 12h   | F30.11    | Initial angle auto-tuning selection  | 0: Disabled<br>1: Enabled             | 0-1         | 0       | -    | U16       | During operation  | Immediately    |
| 15h   | F30.14    | Friction force auto-tuning selection | 0: Disabled<br>1: Enabled             | 0-1         | 0       | -    | U16       | During operation  | Immediately    |
| 22h   | F30.21    | AI self-regulating enable            | 0: Inactive<br>1: AI1 self-regulating | 0-1         | 0       | -    | U16       | At stop           | Immediately    |
| 32h   | F30.31    | VDI1-16 functions                    | -                                     | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |
| 33h   | F30.32    | VDI17-32 functions                   | -                                     | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |
| 34h   | F30.33    | VDI33-48 functions                   | -                                     | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |
| 35h   | F30.34    | VDI49-64 functions                   | -                                     | 0-65535     | 0       | -    | U16       | During operation  | Immediately    |

|     |        |                    |   |         |   |   |     |           |             |
|-----|--------|--------------------|---|---------|---|---|-----|-----------|-------------|
| 36h | F30.35 | VDO1-16 functions  | - | 0-65535 | 0 | - | U16 | Read only | Immediately |
| 37h | F30.36 | VDO17-32 functions | - | 0-65535 | 0 | - | U16 | Read only | Immediately |

## 8.20 Reset Parameters (2031h/F31)

| Index | Parameter | Name                        | Options                                                                                                                                            | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | F31.00    | Fault reset                 | 0: Inactive<br>1: Reset                                                                                                                            | 0-1         | 0       | -    | U16       | At stop           | Immediately    |
| 02h   | F31.01    | Software reset              | 0: Inactive<br>1: Reset                                                                                                                            | 0-2         | 0       | -    | U16       | At stop           | Immediately    |
| 03h   | F31.02    | Parameter initialization    | 0: Inactive<br>1: Restore default settings of parameters<br>2: Restore default settings of the object dictionary                                   | 0-2         | 0       | -    | U16       | At stop           | Immediately    |
| 05h   | F31.04    | Fault record initialization | 0: Inactive<br>1: Fault record clearing                                                                                                            | 0-1         | 0       | -    | U16       | At stop           | Immediately    |
| 11h   | F31.10    | Encoder data reset          | 0: Inactive<br>1: Read encoder<br>2: Write encoder<br>3: Reset encoder fault<br>4: Reset encoder fault and multi-turn data<br>16: Operation failed | 0-31        | 0       | -    | U16       | At stop           | Immediately    |

### NOTICE

- For details about parameters above, refer to section 8.24.7 "Group F31".

## 8.21 Running Monitoring Parameters (2040h/U40)

| Index | Parameter | Name            | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-----------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | U40.00    | Speed reference | -       | -9000-9000  | 0       | rpm  | I16       | Read only         | Immediately    |
| 02h   | U40.01    | Speed feedback  | -       | -9000-9000  | 0       | rpm  | I16       | Read only         | Immediately    |

| Index | Parameter | Name                                        | Options | Value Range          | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|---------------------------------------------|---------|----------------------|---------|---------------------|-----------|-------------------|----------------|
| 03h   | U40.02    | Torque reference                            | -       | -4000-4000           | 0       | 0.1%                | I16       | Read only         | Immediately    |
| 04h   | U40.03    | Torque feedback                             | -       | -4000-4000           | 0       | 0.1%                | I16       | Read only         | Immediately    |
| 05h   | U40.04    | DI status                                   | -       | 0-65535              | 0       | -                   | U16       | Read only         | Immediately    |
| 06h   | U40.05    | DO status                                   | -       | 0-65535              | 0       | -                   | U16       | Read only         | Immediately    |
| 07h   | U40.06    | Bus voltage                                 | -       | 0-9000               | 0       | 0.1V                | U16       | Read only         | Immediately    |
| 08h   | U40.07    | Average load ratio                          | -       | 0-4000               | 0       | 0.1%                | U16       | Read only         | Immediately    |
| 09h   | U40.08    | Electrical angle                            | -       | 0-36000              | 0       | 0.01°               | U16       | Read only         | Immediately    |
| 0Ah   | U40.09    | Mechanical angle                            | -       | 0-36000              | 0       | 0.01°               | U16       | Read only         | Immediately    |
| 0Bh   | U40.0A    | AI1 input voltage                           | -       | -1200-1200           | 0       | 0.01V               | I16       | Read only         | Immediately    |
| 0Dh   | U40.0C    | RMS value of phase current                  | -       | -9000-9000           | 0       | 0.1A                | I16       | Read only         | Immediately    |
| 0Eh   | U40.0D    | High-accuracy AI input voltage              | -       | $-2^{31}-(2^{31}-1)$ | 0       | 0.001mV             | I32       | Read only         | Immediately    |
| 10h   | U40.0F    | Current AI input current                    | -       | 0-40000              | 0       | 0.001mA             | U16       | Read only         | Immediately    |
| 11h   | U40.10    | Position deviation counter                  | -       | $-2^{31}-(2^{31}-1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 13h   | U40.12    | Number of input pulses                      | -       | $-2^{31}-(2^{31}-1)$ | 0       | Unit in application | I32       | Read only         | Immediately    |
| 15h   | U40.14    | Absolute position reference                 | -       | $-2^{31}-(2^{31}-1)$ | 0       | Unit in application | I32       | Read only         | Immediately    |
| 17h   | U40.16    | Absolute position feedback (reference unit) | -       | $-2^{31}-(2^{31}-1)$ | 0       | Unit in application | I32       | Read only         | Immediately    |
| 19h   | U40.18    | Absolute position feedback (encoder unit)   | -       | $-2^{31}-(2^{31}-1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 18h   | U40.1A    | Absolute position feedback (encoder unit)   | -       | $-2^{31}-(2^{31}-1)$ | 0       | P                   | I32       | Read only         | Immediately    |



| Index | Parameter | Name                                                              | Options | Value Range              | Default | Unit                | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------------------------------------|---------|--------------------------|---------|---------------------|-----------|-------------------|----------------|
| 1Dh   | U40.1C    | Encoder single-turn data                                          | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 1Fh   | U40.1E    | Encoder multi-turn position data                                  | -       | 0-65535                  | 0       | Rev                 | U16       | Read only         | Immediately    |
| 20h   | U40.1F    | Encoder initial angle                                             | -       | 0-36000                  | 0       | 0.01°               | U16       | Read only         | Immediately    |
| 21h   | U40.20    | Encoder multi-turn data (low 32 bits)                             | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 23h   | U40.22    | Encoder multi-turn data (high 32 bits)                            | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 25h   | U40.24    | Absolute position feedback (encoder unit) (low 32 bits)           | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 27h   | U40.26    | Absolute position feedback (encoder unit) (high 32 bits)          | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 29h   | U40.28    | Position feedback in rotation mode (reference unit) (low 32 bits) | -       | $-2^{31} - (2^{31} - 1)$ | 0       | Unit in application | I32       | Read only         | Immediately    |
| 2Bh   | U40.2A    | Position feedback in rotation mode (encoder unit) (low 32 bits)   | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 2Dh   | U40.2C    | Position feedback in rotation mode (encoder unit) (high 32 bits)  | -       | $-2^{31} - (2^{31} - 1)$ | 0       | P                   | I32       | Read only         | Immediately    |
| 30h   | U40.2F    | Online inertia auto-tuning value                                  | -       | 0-12000                  | 0       | %                   | U16       | Read only         | Immediately    |
| 31h   | U40.30    | IGBT temperature                                                  | -       | -9000-9000               | 0       | 0.1°C               | I16       | Read only         | Immediately    |
| 32h   | U40.31    | Motor temperature                                                 | -       | -9000-9000               | 0       | 0.1°C               | I16       | Read only         | Immediately    |
| 33h   | U40.32    | Encoder temperature                                               | -       | -9000-9000               | 0       | 0.1°C               | I16       | Read only         | Immediately    |
| 34h   | U40.33    | Ambient temperature                                               | -       | -9000-9000               | 0       | 0.1°C               | I16       | Read only         | Immediately    |
| 35h   | U40.34    | Offline inertia auto-tuning value                                 | -       | 0-12000                  | 0       | %                   | U16       | Read only         | Immediately    |
| 36h   | U40.35    | Control circuit voltage                                           | -       | 0-9000                   | 0       | 0.1V                | U16       | Read only         | Immediately    |

| Index | Parameter | Name                                   | Options                                                                                                                                                                                                                                           | Value Range              | Default | Unit   | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------|--------|-----------|-------------------|----------------|
| 37h   | U40.36    | Instantaneous value in phase U current | -                                                                                                                                                                                                                                                 | $-2^{31} - (2^{31} - 1)$ | 0       | 0.001A | I32       | Read only         | Immediately    |
| 39h   | U40.38    | Instantaneous value in phase V current | -                                                                                                                                                                                                                                                 | $-2^{31} - (2^{31} - 1)$ | 0       | 0.001A | I32       | Read only         | Immediately    |
| 3Fh   | U40.3E    | Drive accumulated heat                 | -                                                                                                                                                                                                                                                 | 0-2000                   | 0       | 0.1%   | U16       | Read only         | Immediately    |
| 40h   | U40.3F    | Motor accumulated heat                 | -                                                                                                                                                                                                                                                 | 0-2000                   | 0       | 0.1%   | U16       | Read only         | Immediately    |
| 43h   | U40.42    | Drive NotRdy state                     | Bit00: Control circuit error<br>Bit01: Main circuit power input error<br>Bit02: Bus undervoltage<br>Bit03: Soft start failure<br>Bit04: Encoder initialization not done<br>Bit05: Short-circuit to ground failed<br>Bit06: Other                  | 0-255                    | 0       | -      | U16       | Read only         | Immediately    |
| 44h   | U40.43    | Motor running failure reason           | 0: No reason<br>1: Servo not ready<br>2: No servo running command<br>3: Overtravel disabled<br>4: Small torque limit<br>5: Small position reference<br>6: Small speed reference<br>7: Small torque reference<br>8: Small speed limit<br>9: Others | 0-65535                  | 0       | -      | U16       | Read only         | Immediately    |
| 51h   | U40.50    | Encoder initial position               | -                                                                                                                                                                                                                                                 | $0 - (2^{32} - 1)$       | 0       | Unit   | U32       | Read only         | Immediately    |

## NOTICE

- For details about parameters above, refer to section 8.24.8 "Group U40".

## 8.22 Status Monitoring Parameters (2041h/U41)

| Index | Parameter | Name                                                        | Options                                                                    | Value Range    | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|-------------------------------------------------------------|----------------------------------------------------------------------------|----------------|---------|------|-----------|-------------------|----------------|
| 01h   | U41.00    | MCU system status                                           | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 02h   | U41.01    | MCU fault state                                             | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 03h   | U41.02    | FPGA system status                                          | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 04h   | U41.03    | FPGA fault state                                            | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 05h   | U41.04    | Encoder system status                                       | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 06h   | U41.05    | Encoder fault state                                         | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 07h   | U41.06    | Group number of abnormal parameter                          | -                                                                          | 0-255          | 0       | -    | U16       | Read only         | Immediately    |
| 08h   | U41.07    | Offset of the abnormal parameter within the parameter group | -                                                                          | 0-255          | 0       | -    | U16       | Read only         | Immediately    |
| 09h   | U41.08    | Position planning operation group number                    | -                                                                          | 0-16           | 0       | -    | U16       | Read only         | Immediately    |
| 0Ah   | U41.09    | Speed planning operation group number                       | -                                                                          | 0-16           | 0       | -    | U16       | Read only         | Immediately    |
| 0Bh   | U41.0A    | Servo Status                                                | 0: Servo not ready<br>1: Servo ready<br>2: Servo running<br>3: Servo fault | 0-3            | 0       | -    | U16       | Read only         | Immediately    |
| 0Ch   | U41.0B    | Servo running mode                                          | -                                                                          | 0-9            | 0       | -    | U16       | Read only         | Immediately    |
| 0Dh   | U41.0C    | Servo running time                                          | -                                                                          | $0-(2^{32}-1)$ | 0       | 0.1s | U32       | Read only         | Immediately    |
| 11h   | U41.10    | Communication encoder status bit                            | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |
| 12h   | U41.11    | Communication encoder fault bit                             | -                                                                          | 0-65535        | 0       | -    | U16       | Read only         | Immediately    |

## 8.23 Version Parameters (2042h/U42)

| Index | Parameter | Name                                   | Options | Value Range | Default | Unit | Data Type | Modification Mode | Effective Time |
|-------|-----------|----------------------------------------|---------|-------------|---------|------|-----------|-------------------|----------------|
| 01h   | U42.00    | ARM version                            | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 02h   | U42.01    | FPGA version                           | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 03h   | U42.02    | Encoder version                        | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 04h   | U42.03    | ARM-based machine                      | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 05h   | U42.04    | FPGA-based machine                     | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 06h   | U42.05    | Internal software version              | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 07h   | U42.06    | Modbus version                         | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 10h   | U42.0F    | Test software version                  | -       | 0-65535     | 0       | 0.01 | U16       | Read only         | Immediately    |
| 11h   | U42.10    | Drive model                            | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 12h   | U42.11    | Motor model                            | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 13h   | U42.12    | Encoder model                          | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 14h   | U42.13    | Power supply unit model identification | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 15h   | U42.14    | Inverter model identification 1        | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 16h   | U42.15    | Inverter model identification 2        | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 17h   | U42.16    | Servo version                          | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |
| 18h   | U42.17    | Servo chip type                        | -       | 0-65535     | 0       | -    | U16       | Read only         | Immediately    |

## 8.24 Description of Parameters

### 8.24.1 Group C00

#### C00.05: Stiffness level

- Defines the stiffness level of the servo system. The higher the stiffness level, the stronger the gains and the quicker the response will be. But an excessively high stiffness level will cause vibration. The setpoint 0 indicates the lowest stiffness and 41 indicates the highest stiffness.

#### C00.06: Load inertia ratio

- Defines the mechanical load inertia ratio relative to the motor moment of inertia.
- When C00.06 is set to 0, it indicates the motor carries no load; if it is set to 1.00, it indicates the mechanical load inertia is the same as the motor moment of inertia.
- When the value of C00.06 is equal to the actual inertia ratio, the value of speed loop gain can represent the maximum follow-up frequency of actual speed loop.

### 8.24.2 Group C01

#### C01.00: 1st position loop gain

- Defines the proportional gain of the position loop.
- This parameter determines the responsiveness of the position loop. A high setpoint shortens the positioning time. Note that an excessively high setpoint may cause vibration.
- The 1st gain set includes C01.00, C01.01, C01.02, and C01.03.

#### C01.01: 1st speed loop gain

- Defines the speed loop proportional gain.
- This parameter determines the responsiveness of the speed loop. The higher the setpoint, the faster the speed loop response is. Note that an excessively high setpoint may cause vibration.
- In the position control mode, the position loop gain must be increased together with the speed loop gain.

#### C01.02: 1st speed loop integral time

- Defines the speed loop integral time constant.
- The lower the setpoint, the better the integral action, and the quicker will the deviation value be close to 0.
- There is no integral action when C01.02 is set to 512.00 ms

#### C01.08: 2nd position loop gain

- Defines the 2nd gain of the position loop.
- The 2nd gain set includes C01.08, C01.09, C01.0A, and C01.0B.
- For details about gain switchover, see section 7.5 "Gain Switchover".

#### C01.0B: 2nd torque reference filter cutoff frequency

- Defines the torque reference filter time constant.

- Low-pass filtering of torque references helps to smoothen torque references and reduce vibration.
- Pay attention to the responsiveness during setting as an excessively high setpoint lowers down the responsiveness.

### NOTICE

- The servo drive offers two low-pass filters for torque references. By default, the 1st filter is used.
- Gain switchover can be used in the position or speed control mode. Once certain conditions are satisfied, the servo drive can switch to filter 2.

#### C01.11: Cutoff frequency of speed feedback low-pass filter

- Defines the cutoff frequency for first-order low-pass filtering on the speed feedback.
- The lower the setpoint, the weaker the speed feedback fluctuation, and the longer the feedback delay will be.
- Setting this parameter to 8000 Hz negates the filtering effect.

#### C01.12: Speed feedback overlapping average filter time constant

- Defines the moving average filtering times for speed feedback.
- The higher the setpoint, the weaker the speed feedback fluctuation, but the longer the feedback delay will be.
- When C01.12 is set to a value higher than 0, C01.11 (Cutoff frequency of speed feedback low-pass filter) is invalid.

#### C01.13: Speed feedforward source

- Defines the source of the speed loop feedforward signal.
- In the position control mode, the speed feedforward control can improve the position reference responsiveness.

| Setpoint | Speed Feedforward Source | Remarks                                                                                                                                                                                                                                                                                      |
|----------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | No feedforward           | -                                                                                                                                                                                                                                                                                            |
| 1        | Internal reference       | The speed corresponding to the position reference (encoder unit) is defined as the speed feedforward source.                                                                                                                                                                                 |
| 2        | Model tracking           | Model tracking control can improve the responsiveness and shorten the positioning time.<br>It is only available in the position control mode.<br>It must be used with C02.00. When C02.00 is set to 1, the speed feedforward is sourced from the speed feedforward output of model tracking. |

C01.14: Speed feedforward percentage

- In the position control mode, speed feedforward is the value of C01.14 multiplied by the speed feedforward signal, which is part of the speed reference. Increasing the setpoint improves the responsiveness to position references and reduces the position deviation during operation at a constant speed.
- Set C01.15 to a fixed value first, and then gradually increase the value of C01.14 from 0 to a certain setpoint at which speed feedforward achieves the desired effect.
- Adjust C01.15 and C01.14 repeatedly until a balanced setting is achieved.
- For the speed feedforward function and speed feedforward signal selection, see C01.13 (Speed feedforward source selection).

C01.15: Speed feedforward filter cutoff frequency

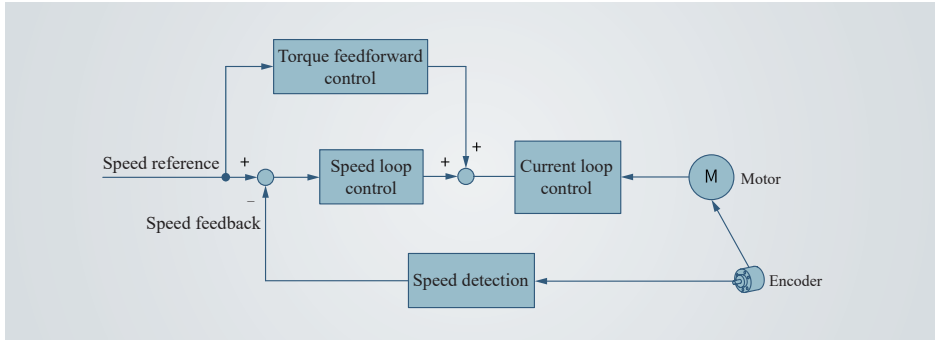
- Defines the speed feedforward smoothing filter time.

C01.16: Torque feedforward source

- Defines whether to enable the internal torque feedforward function in a non-torque control mode.
- The torque feedforward function can improve the torque reference responsiveness and reduce the position deviation during operation at constant acceleration/deceleration rate.

| Setpoint | Torque Feedforward Source | Remarks                                                                                                                                                                                                                                        |
|----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | No feedforward            | -                                                                                                                                                                                                                                              |
| 1        | Internal reference        | The torque feedforward signal source is the speed reference. In the position control mode, the speed reference is output from the position controller. In the speed control mode, the speed reference is output from the user speed reference. |
| 2        | Model tracking            | It must be used with C02.00. When C02.00 is set to 1, the torque feedforward is sourced from the torque feedforward output of model tracking.                                                                                                  |
| 5        | Communication             | In CSP, 60B1h is used as the source of the external torque feedforward signal. Bit 6 of 607Eh can specify the polarity of the torque feedforward signal (60B1h).                                                                               |

- Torque feedforward parameters include C01.17 (Torque feedforward percentage) and C01.18 (Torque feedforward cutoff frequency).
- In a non-torque control mode, the control block diagram of torque feedforward is as follows:



**Figure 8-1 Torque feedforward control**

#### C01.17: Torque feedforward percentage

- In control modes other than torque control, torque feedforward is the product of torque feedforward signal multiplied by C01.17 and is part of the torque reference. Increasing the setpoint improves the responsiveness to variable speed references and position references and reduces the position deviation during operation at a constant speed.

#### C01.18: Torque feedforward filter cutoff frequency

- Defines the filter time constant of torque feedforward.

#### C01.1B: PDFF control coefficient

- Defines the control method of the speed loop.
- When the setpoint is 100.0, PI control (default control mode of the speed loop) is applied to the speed loop, which features fast dynamic response.
- When the setpoint is 0.0, speed loop integral action is enhanced, which filters out low-frequency interference but also slows down the dynamic response.
- C01.1B can be used to keep a good responsiveness of the speed loop, with the anti-interference capacity in low-frequency bands improved and the speed feedback overshoot not increased.

#### C01.30: Adaptive notch mode

Setpoint:

- 0: Adaptive notch not updated
- 1: One adaptive notch activated (3rd notch)
- 2: Two adaptive notches activated (3rd and 4th notches)
- 3: Adaptive notch cleared, values of the 3rd and 4th notches restored to default settings
- 4: Resonance point tested only, displayed in C01.31, C01.32, and C01.33

Description:

- Defines the operation mode of the adaptive notch.



C01.38: Gain switchover mode

| Setpoint | Gain Switchover Condition   | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | Fixed to the 1st gain set   | The 1st gain set applies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1        | DI switchover               | Gains are switched through bit 26 of 60FE.<br>Bit 26 signal inactive: 1st gain set<br>Bit 26 signal active: 2nd gain set<br>If the bit 26 signal cannot be allocated to a DI terminal, the 1st gain set applies.                                                                                                                                                                                                                                                                                                            |
| 2        | DI P-PI switchover          | Gains are switched through bit 26 of 60FE.<br>Bit 26 signal inactive: 1st gain set<br>Bit 26 signal active: 2nd gain set (The 2nd speed loop integral (C01.0A) is forced to be 512 ms.)<br>If the bit 26 signal cannot be allocated to a DI terminal, the 1st gain set applies.                                                                                                                                                                                                                                             |
| 3        | Torque reference            | When the absolute value of the torque reference exceeds (threshold + loop width, %) in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the absolute value of the torque reference is less than (threshold – loop width, %) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.                                                                                                                                                          |
| 4        | Speed reference             | When the absolute value of the speed reference exceeds (threshold + loop width, rpm) in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the absolute value of the speed reference is less than (threshold – loop width, rpm) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.                                                                                                                                                        |
| 5        | Speed feedback              | It is valid only in the position control mode.<br>When the absolute value of the actual speed exceeds (threshold + loop width, rpm) in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the absolute value of the actual speed is less than (threshold – loop width, rpm) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.<br>The 1st gain set applies when the drive is not in the position control mode.                            |
| 6        | Speed reference change rate | It is valid only in non-speed control modes.<br>When the absolute value of the change rate in the speed reference exceeds (threshold + loop width, 10 rpm/s) in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the absolute value of the change rate in the speed reference is less than (threshold – loop width, 10 rpm/s) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.<br>The 1st gain set applies in the speed control mode. |

| Setpoint | Gain Switchover Condition | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7        | Position deviation        | When the absolute value of the position deviation exceeds (threshold + loop width, encoder unit) in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the absolute value of the position deviation is less than (threshold + loop width, encoder unit) and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.<br>The 1st gain set applies when the drive is not in the position control mode. |
| 8        | Position reference        | It is valid only in the position control mode.<br>When the position reference is not 0 in the last 1st gain set, the drive switches to the 2nd gain set.<br>When the position reference is 0 and this status lasts within the delay (C01.39) in the last 2nd gain set, the drive returns to the 1st gain set.<br>The 1st gain set applies when the drive is not in the position control mode.                                                                                |

#### C01.39: Gain switchover time

Defines the duration when the drive switches from the 2nd gain set to the 1st gain set.

#### C01.3A: Gain switchover threshold

- Defines the gain switchover threshold.
- Gain switchover is affected by both the threshold and the loop width, as defined by C01.38. The unit of gain switchover threshold varies with the switchover condition.
- Set C01.3A to a value greater than or equal to C01.3B. If C01.3A is set to a value less than C01.3B, the servo drive sets C01.3A to the same value as C01.3B.

#### C01.3B: Gain switchover loop width

- Defines the gain switchover loop width.
- Gain switchover is affected by both the threshold and the loop width. The unit of gain switchover threshold varies with the switchover condition.
- Set C01.3A to a value greater than or equal to C01.3B. If C01.3A is set to a value less than C01.3B, the servo drive sets C01.3A to the same value as C01.3B.

#### C01.40: Frequency of the 1st notch

- Defines the center frequency of the notch, which is the mechanical resonance frequency.
- In the torque control mode, setting the notch frequency to 8000 Hz deactivates the notch function.

#### C01.41: Width level of the 1st notch

- Defines the width level of the notch. Use the default value in general cases.
- Width level is the ratio of the notch width to the notch center frequency.

#### C01.42: Depth level of the 1st notch

- Defines the depth level of the notch.

- The depth level of the notch is the ratio between the input to the output at the notch center frequency.
- The higher the setpoint, the lower the notch depth and the weaker the mechanical resonance suppression will be. Note that an excessively high setpoint may cause system instability.
- For the use of notch, see 7.14 "Vibration Suppression".

#### C01.45: Depth level of the 2nd notch

- Description of the 2nd notch parameters is the same as that of the 1st notch parameters.

### NOTICE

- The 1st and 2nd notches can be set manually or configured as adaptive notches (C01.30 = 1 or 2). In this case, the parameters are automatically set by the drive, while the other three notches can be set manually.

## 8.24.3 Group C03

### C03.21: Speed reference

- It is the speed reference in the local speed mode, which is invalid in EtherCAT mode.

### C03.22: Acceleration rate

- It is the acceleration ramp time of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

### C03.24: Deceleration rate

- It is the deceleration ramp time of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

### C03.27: Internal positive speed limit

- It is the PL of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

### C03.28: Internal negative speed limit

- It is the NL of the speed reference in the local speed mode, which is invalid in EtherCAT mode.

### C03.43: Internal positive torque limit

- It is valid only in the local torque mode. For torque limit in EtherCAT mode, use 60E0h/60E1h/6072h. Use the torque limit with caution as an excessively low limit value may lead to insufficient motor torque output.
- If the setpoint exceeds the maximum torque of the servo motor and servo drive, the actual torque is limited to the maximum torque of the servo motor and servo drive.

### C03.44: Internal negative torque limit

- It is valid only in the local torque mode. For torque limit in EtherCAT mode, use 60E0h/60E1h/6072h. Use the torque limit with caution as an excessively low limit value may lead to insufficient motor torque output.

- If the setpoint exceeds the maximum torque of the servo motor and servo drive, the actual torque is limited to the maximum torque of the servo motor and servo drive.

#### C03.47: Positive speed limit in torque mode

- It is valid only in the local torque mode. Use 607F for the speed limit in the EtherCAT, CST, and PT modes.

#### C03.48: Negative speed limit in torque mode

- It is valid only in the local torque mode. Use 607F for the speed limit in the EtherCAT, CST, and PT modes.

#### C03.4B: Invalid value for torque reached

- The torque reached function is used to judge whether the actual torque reference reaches the range of the valid value for torque reached. If yes, the servo drive outputs the corresponding flag (bit 10 of the status word) to the host controller.

A: Actual torque reference (U40.02)

B: Base value for torque reach (C03.49)

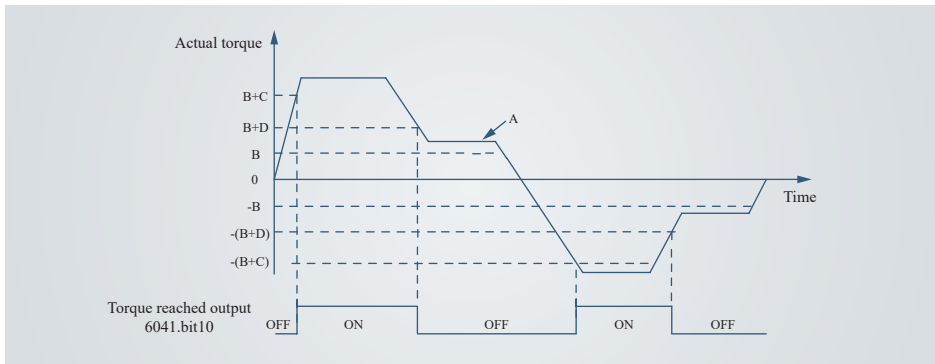
C: Valid value for torque reach (C03.4A)

D: Invalid value for torque reach (C03.4B)

C and D are offsets on the basis of B.

The torque reach signal is activated only when the actual torque reference meets the condition:  $|A| \geq B + C$ . Otherwise, the torque reach signal remains inactive.

The torque reach signal is deactivated only when the actual torque reference meets the condition:  $|A| < B + D$ .



## 8.24.4 Group C06

### C06.04: Input phase loss detection

Servo drives support three-phase 380 V power supplies. When voltage fluctuation or phase loss occurs on the power supply, power input phase loss protection will be triggered by the servo drive based on the setting of C06.04.

- C06.04 = 0: The servo drive reports Er81.0 (Phase loss fault) when the servo drive is set to 3 kW.
- C06.04 = 1: The servo drive does not report Er81.0 (Phase loss fault) when the servo drive is set to 3 kW, with deration of 80%.

### C06.11: Motor overload protection threshold

- Determines the motor overload duration before Er41.0 (Motor overload) is reported.
- You can change the setpoint to advance or delay the time when overload protection is triggered based on the motor temperature. The setpoint 50% indicates the time is cut by half; 150% indicates the time is prolonged by 50%.
- Set this parameter based on the actual temperature of the motor.

### C06.20: Protection from out of control

- Sets whether to enable the runaway protection function.

## 8.24.5 Group C0A

### C0A.09: Commissioning software communication baud rate

- Defines the communication rate between the servo drive and the host controller.
- The baud rate set in the servo drive must be the same as that in the host controller. Otherwise, communication will fail.

### C0A.0A: Commissioning software communication format

- Defines the data check mode between the servo drive and the host controller during communication.
- The data format of the servo drive and the host controller must be the same; otherwise, the communication fails.

## 8.24.6 Group R21

### R21.00: Drive model

Setpoint:

2: 200RS

3: 400RS

5: 750RS

6: 1000RS

Description:

- Sets the SN of the servo drive. The following table lists the servo drive SNs.

| Setpoint | Servo Drive SN | Remarks                                                                                                                                                                                           |
|----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | 200RS          | The rated drive power is 0.2 kW. The main circuit inputs single-phase 220 V.                                                                                                                      |
| 3        | 400RS          | The rated drive power is 0.4 kW. The main circuit inputs single-phase 220 V.                                                                                                                      |
| 5        | 750RS          | The rated drive power is 0.75 kW. The main circuit inputs single-phase 220 V.                                                                                                                     |
| 6        | 1000RS         | The rated drive power is 1.0 kW. The main circuit inputs single-phase or three-phase 220 V.<br>(The main circuit of the servo drive supports single-phase 220 V power supplies without derating.) |

If the voltage input to the main circuit of the servo drive does not comply with the preceding specifications, a fault or damage occurs.

### 8.24.7 Group F31

#### F31.00: Fault reset

- Defines whether to enable fault reset.

| Setpoint | Function     | Remarks                                                                                                                                                                                                                                                                                |
|----------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | No operation | -                                                                                                                                                                                                                                                                                      |
| 1        | Enable       | When a No.1 or No.2 resettable fault occurs, you can enable the fault reset function in the non-operational state after rectifying the fault cause and stopping the keypad from displaying the fault.<br>When a No.3 warning occurs, you can enable the fault reset function directly. |

- For fault classification, see section 10.1.3 "List of faults and alarms".
- The fault reset function, once enabled, stops the keypad from displaying the fault only. It does not activate modifications made on parameters.
- This function is not applicable to non-resettable faults. Use this function with caution in cases where the fault causes are not rectified.

#### F31.01 Software reset

- Defines whether to enable fault reset.

| Setpoint | Function     | Remarks                                                                                                                                                                      |
|----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0        | No operation | -                                                                                                                                                                            |
| 1        | Enable       | Programs in the drive are reset automatically (similar to the program reset upon power-on) after the software reset function is enabled, without the need for a power cycle. |

- Software reset conditions: The servo drive is disabled, and there is no non-resettable fault such as No.1 fault.

F31.10: Encoder data reset

- The absolute position saved by the encoder changes abruptly after multi-turn data reset. In this case, perform mechanical homing.

**8.24.8 Group U40**

U40.00: Speed reference

- Indicates the present speed reference (accurate to 1 RPM) of the drive in the position and speed control modes.

U40.01: Speed feedback

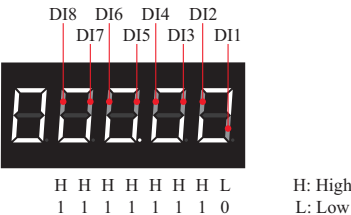
- Indicates the actual motor speed after round-off, which is accurate to 1 rpm.
- This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

U40.02: Actual torque reference

- Indicates the present torque reference (accurate to 0.1%). The value 100.0% corresponds to the rated torque of the motor.

U40.04: DI status

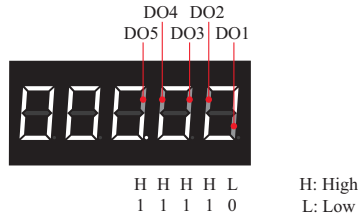
- Indicates the level status of eight DIs without filtering.
- Upper LED segments ON: high level (indicated by "1")  
Lower LED segments ON: low level (indicated by "0") In cases where DI1 is low level and DI2 to DI8 are high level, the corresponding binary value is 11111110, and the value of U40.04 read in the software tool is 0xFFFE.
- The keypad displays as follows:



U40.05: DO state

- Indicates the level status of five DOs without filtering.
- Upper LED segments ON: high level (indicated by "1")  
Lower LED segments ON: low level (indicated by "0") In cases where DO1 is low level and DO2 to DO5 are high level, the corresponding binary value is 111110, and the value of U40.05 read in the software tool is 0xFFFE.

- The keypad displays as follows:



#### U40.06: Bus voltage

- Indicates the DC bus voltage of the main circuit input voltage after rectification, which is accurate to 0.1 V.

#### U40.07: Average load ratio

- Indicates the percentage of the average load torque to the rated torque of the motor, which is accurate to 0.1%. The value 100.0% corresponds to the rated torque of the motor.

#### U40.08: Electrical angle

- Indicates the present electrical angle of the motor, which is accurate to 0.1°.
- The electrical angle variation range is  $\pm 360.0^\circ$  when the motor rotates.
- If the motor has four pairs of poles, each revolution generates four rounds of angle changes from  $0^\circ$  to  $359.9^\circ$ .
- Similarly, if the motor has five pairs of poles, each revolution generates five rounds of angle changes from  $0^\circ$  to  $359.9^\circ$ .

#### U40.09: Mechanical angle

- Indicates present mechanical angle (encoder unit) of the motor. The value 0 indicates that the mechanical angle is  $0^\circ$ .

#### U40.0C: RMS value of phase current

- Indicates the RMS value of the phase current of the servo motor, which is accurate to 0.1 A.

#### U40.10: Position deviation counter

- Counts the position pulses fed back by the encoder in any control mode.
- This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

#### U40.30: IGBT temperature

- Indicates the temperature of the module inside the servo drive, which can be used as a reference for estimating the actual temperature of the servo drive.